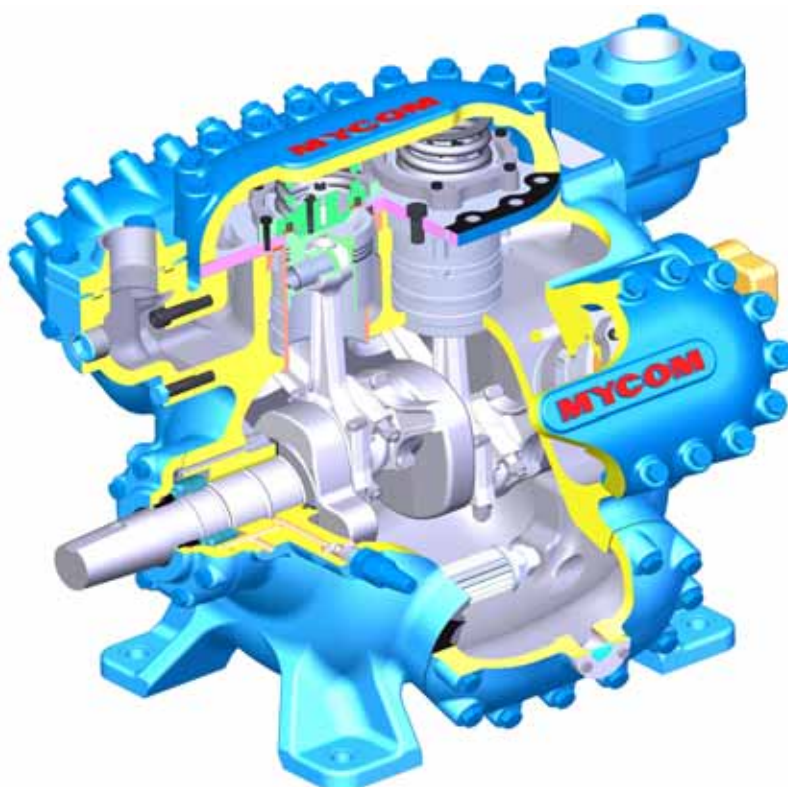


MYCOM

Model 6HK Reciprocating Compressor INSTRUCTION MANUAL (Maintenance)



Read this instruction manual carefully before commencing compressor operation or undertaking inspection or maintenance service.

MAYEKAWA MFG. CO., LTD.
Tokyo, Japan

Cat.No.0210R8CJ-CD411C2005.03

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Safety Information

This manual has been prepared to protect engineers, system operators and maintenance personnel from any potential risks during maintenance of Model 6HK Reciprocating Compressors and to assure effective utilization of the unit. Before commencing disassembly and reassembly work on the compressor, read carefully and understand the contents of this Instruction Manual (Maintenance), which explains the structure of the Model 6HK Reciprocating Compressor and the appropriate work methods.

If any service work is performed which disregards the instructions given in this manual, there is a risk of accident. Most accidents and problems encountered during maintenance are a result of negligence in following basic precautions, insufficient understanding of and failure to observe proper operating principles and procedures and failure to implement appropriate inspection and maintenance procedures and precautions.

Safety and signal warnings provided in this instruction manual are classified into the following three categories.



“DANGER” indicates a hazardous situation in which failure to observe all safety precautions will lead to death or serious injury and major damage to the compressor system.



“WARNING” indicates a hazardous situation in which failure to observe all safety precautions may lead to death or injury and major damage to the compressor



“CAUTION” indicates a hazardous situation in which failure to observe all safety precautions could lead to injury or damage to the compressor

The safety and signal warnings contained in this instruction manual establish the minimum level of safety required to perform maintenance work on the compressor. System operators and staff involved in maintenance service are advised to implement at their own responsibility any further safety activities in line with the particular environment or the compressor system.

1. Safety

1.1. Basic Safety Warnings

The 6HK Reciprocating Compressor Instruction Manual has been prepared to assure maximum effective operation of the compressor as well as to ensure the safety of those involved in overhauling and maintenance of the compressor and related parts. Read this manual carefully in order to familiarize yourself with the structure and procedures before undertaking disassembly and reassembly of this compressor. Failure to follow the instructions given in this manual may result in possible personal injury and/or serious compressor damage.



Should this compressor be operated carelessly and in disregard of the instructions given in this manual, death or serious injury may result. This manual must therefore be kept in a convenient, easily-accessible location near the system and should be studied periodically by those working with the compressor.

- Keep this manual in a convenient, easily-accessible location near the compressor and study periodically and frequently.
- Do not commence overhauling or maintenance work on the compressor before reading and fully understanding this instruction manual.
- Prior to commencing any inspection or maintenance work on the compressor, read the safety warnings provided.
- Prior to commencing overhauling or maintenance work on the compressor, read the instruction manual to understand and familiarize yourself with all tools used.
- Management of this compressor and related components for the refrigeration system should be strictly controlled and unauthorized persons should not be allowed to touch the equipment (except for the control valves and switches in an emergency).
- If this manual is lost or damaged, a replacement should be obtained immediately from Mayekawa Mfg. Co., Ltd. or the nearest representative in your region.
- If ownership of the compressor passes to another party, this instruction manual or the electronic data file, (FD, MO, CD, etc.) containing this manual should always accompany the compressor.
- Mayekawa Mfg. Co., Ltd. reserves the right to make changes or improvements to its products without notice. It is possible, therefore, that some explanations given in this manual may not apply to a particular machine. If any question exists regarding this instruction manual, contact Mayekawa Mfg. Co., Ltd. or the nearest representative in your region.
- Various warning labels are affixed to the compressor. Do not deface or peel off these warning labels.

1.2 Safety Information

Safety information indicates safety cautions described in this instruction manual. Failure to follow any and all cautionary safety information instructions may result in possible personal injury, death and/or serious compressor damage.

2. Introduction

MYCOM Model 6HK reciprocating compressors are newly designed compressors suitable for higher design pressure and higher operating pressure than that of existing models such as the W, K and L Series. The Model 6HK compressor is, therefore, suitable for operation in a binary refrigeration system using carbon dioxide (CO₂) refrigerant and other environmentally friendly refrigerants in place of conventional HCFC refrigerants such as R-22, R-502, etc. The Model 6HK compressor was developed based on the K Series compressor.

The Model 6HK reciprocating compressor operates at a maximum suction pressure of 1.5MPa. and a maximum discharge pressure of 4.0MPa. Utilization of the compressor in a binary refrigeration system for CO₂ refrigerant or operation with refrigerant of high saturation pressure such as R410A for high temperature heat pump systems is possible.

The 6HK compressor is designed complete with a built in piping system except for the oil filter and oil cooler, providing the superior characteristics of the K series compressor while assuring a neat external appearance and easy maintenance accessibility.

All components and parts are manufactured under a high quality management system and maintained up to assembly, inspection and final delivery inspection to assure an extended operating life.

This manual should be read and reread frequently to ensure maximum familiarity with the operation and maintenance work required to maintain the compressor in top condition.

Should you have any questions or comments regarding the contents of this instruction manual, contact Mayekawa Mfg. Co., Ltd. or the MYCOM representative office or liaison office nearest you.

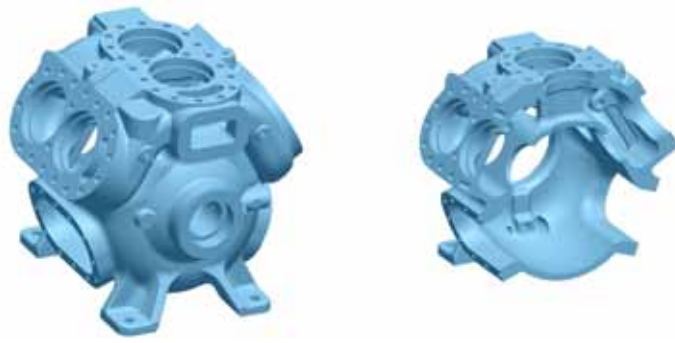
3. Names of Parts and Functions

(Refer to the attached List of Component Parts)

3.1 Crankcase (1)

The crankcase is an essential pressure-tight structural body of the compressor and is designed to secure all component parts. The inside the crankcase consists of two different chambers, one for suction and the other for the crankshaft together with a discharge gas passage enclosed in the strong cast steel structural body.

Taking the characteristic of the Model 6HK compressor operating at higher pressure than conventional reciprocating compressors into consideration, the strength of the crankcase and related component parts are manufactured to withstand high operating pressure.



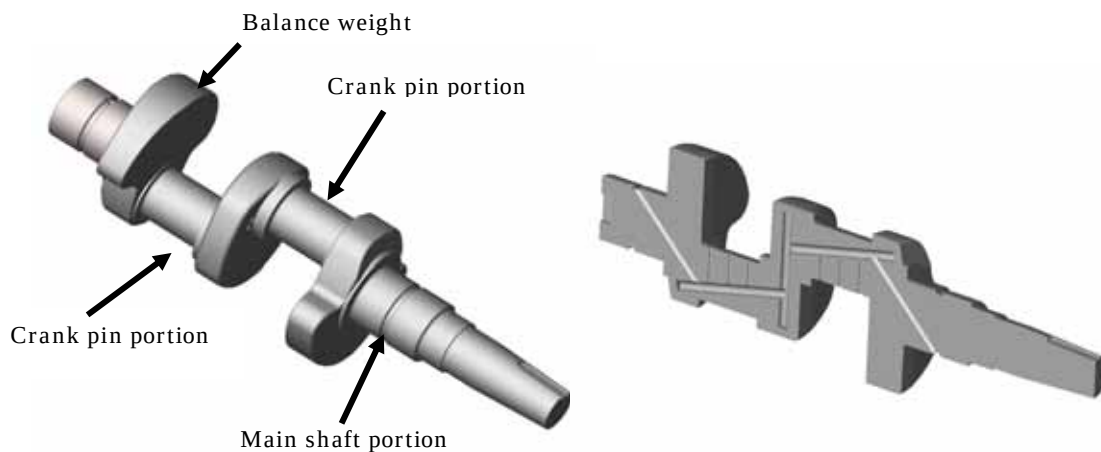
3.2 Crankshaft (2)

The crankshaft is furnished with the crank pins branched from the main shaft to mount the connecting rods. The crank pins are located 180 deg. apart and reciprocate the pistons once every one rotation of the crankshaft to minimize torque difference during the compression process.

The pistons reciprocate together with the connecting rod as the crankshaft rotates. The stroke of the piston is determined by the distance between the crank pin and the shaft portion of the crankshaft, with the stroke of the piston being equal to twice the offset distance (or equal to the distance between the two crank pins).

The crankshaft is bored with an internal oil passage, and lubricating oil sent from the oil pump is supplied to the bearing halves (84), the main shaft at the shaft seal, mechanical shaft seal (32), thrust washers (29-1 & 29-2), etc. through this oil passage.

To minimize the level of vibration due to imbalance of the connecting rods, pistons, etc., the crankshaft is also furnished with a balance weight.



Crank pin portion

3.3 Connecting Rod (77)

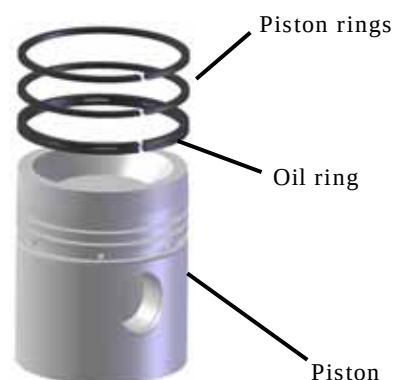
The connecting rod which joins the crankshaft and piston is made of an aluminum alloy taking lower elasticity and maintenance accessibility into consideration. The big end of the connecting rod is a split nut to hold the crank pin from the top and bottom. Roundness is finished after the half cut. Do not change the direction of assembly, otherwise seizure, abrasion, etc. may occur. The big end of the connecting rod is embossed with 3 signal numbers to indicate the proper combination.

Confirm that the numbers are correct before and after assembly. The big end of the connecting rod is furnished with a steel liner, consisting of bearing halves (84). The small end of the connecting rod is lubricated by oil from inside the crankshaft. For this purpose, the upper half of the connecting rod has an oil hole through it and there is an oil groove in the bearing itself.



3.4 Piston (85) and Piston Ring (89~90)

The piston reciprocates inside the cylinder sleeve by rotation of the crankshaft to compress the refrigerant gas. The material of the piston is an aluminum alloy, taking the low inertia force and defacement into consideration. There is a slight gap between the piston and cylinder, from which refrigerant gas could leak. To prevent leakage, two piston rings are fitted to the piston. The piston is also furnished with one oil ring below the two piston rings to control the quantity of oil on the surface of the cylinder sleeve.



3.5 Cylinder Sleeve (61)

The cylinder sleeve is made of special high grade anti-friction cast metal with excellent oil retaining characteristics, because the piston and piston rings moves inside the cylinder sleeve while contacting the internal surface of the cylinder sleeve at extremely high speed. The internal surface of the cylinder sleeve is finished by honing. The outer portion of the cylinder sleeve is finished to accept installation of the unload mechanism, lift pin and cam ring, etc. for capacity control. The top of the cylinder sleeve functions as a seal for the suction valve. Care must be taken to protect the sleeve from scarring during maintenance.



3.6 Head Cover (49 & 50) and Head Springs (117)

The head cover is an essential component protecting the valves and various internal components and is designed for high pressure operating conditions. For a refrigerant with high discharge temperature such as ammonia, a water-cooled jacket type head cover is selected to cool down the head cover by supplying water and prevent overheating and burning. The head cover also functions to secure the head springs. The head springs are installed as a protection mechanism to prevent the discharge plate valves, pistons, etc. from damage due to liquid compression inside the cylinder caused by oil and liquid refrigerant entering the cylinder and instantaneously raising the internal pressure.



3.7 Hand Hole Cover (45)

A hand hole cover is fitted to the lower portion of the compressor crankcase to facilitate disassembly and reassembly. The hand hole cover is furnished with an oil sight glass.



3.8 Suction Valve (71) and Discharge Valve (110)

The suction valve and the discharge valve, also called the ring valve or plate valve, have excellent characteristics against wear and metal fatigue. This ring-like valve can occupy the larger gas passage against the height of the lift and is suitable for operation at high revolution speed and for medium and large-scale compressors. Both suction and discharge valves also serve as check valves, designed to open or close respectively at the suction and discharge stroke only. The suction and discharge springs function to close and secure the ring valves against the valve seat immediately at the end of each cycle.



3.9 Oil Supply Pump and Oil Filter (119)

A built-in oil supply pump fitted to the bearing housing is driven by the crankshaft. The oil supply pump is a gear pump, and the suction side is furnished with an oil filter to prevent compressor damage due to foreign matter contamination. The discharge side of the oil pump is furnished with an external oil filter to remove the smallest foreign particles. The oil pump for the Model 6HK compressor is reversible, but discharges oil in a fixed direction regardless of the direction. The rotating direction of the motor or engine can, therefore, be completely ignored.



Oil Filter

3.10 Oil Cooler

Oil is used in the Model 6HK compressor to lubricate and cool the rotating portions of the compressor. When the heat given off by the rotating portion and heat emission from the surface of the crankcase are equal, oil temperature becomes constant. Oil temperature changes according to the operating conditions, ambient temperature and kind of refrigerant. Oil temperature is maintained at a comparatively low level when the refrigerant is HCFC, but rather high when the refrigerant is NH₃. If the oil temperature becomes excessively high, oil film will not be generated on the bearings, leading to deterioration of the oil.

3.11 Bearing

The main bearing portion of the crankshaft and the big end of the connecting rod are furnished with bearing halves fitted with soft plain metal. The shaft is raised somewhat by an oil film on the shaft and rotates with no contact between the shaft and the bearing, so there is no abrasion. If small foreign particles enter the oil and are sent to the bearing, the shaft can be abraded and scratched. If the crankshaft and bearing are hard against each other, the foreign particles will remain between the shaft and bearing, resulting in acceleration of abnormal abrasion. By using a soft bearing metal becomes buried in the material, protecting the crankshaft from abrasion.



3.12 Shaft Seal (32)

This open type reciprocating compressor is furnished with a shaft seal for the shaft portion extending outside the crankcase. The shaft seal consists of a ring rotating with the shaft, a fixed ring fitted in the shaft seal cover, an O-ring, etc. The surface of the shaft seal is finished by combining cast steel and carbon. A thin oil film generated between the mirror finished rotating ring and the fixed ring and a small amount of oil is exhausted outside the crankcase during operation. Though the shaft seal is cooled by lubricating oil, if the oil flow is interrupted, seizure is unavoidable because the shaft seal rotating portion generates heat. Seizure, breakage or peeling of carbon within a short of time is possible. Should the compressor be left in stop status for an extended period of time, oil inside the seal box and refrigerant will leak out.



4. Disassembly



DANGER

- Before commencing overhaul and maintenance work on the compressor, turn off the main motor power supply and the control power supply switch. Special care must be taken to ensure that the power supply is never connected or turned on when working on the system. If energized during service work, the compressor may start up, posing a threat of electric shock or physical harm to anyone in the vicinity of the rotating shaft.



warning

- * Before commencing overhaul and maintenance work, turn off the main motor power supply, the control power supply and the heater switch. If energized during service work, there is a risk of electric shock, burns or other physical harm.
- * Special care must be taken when turning the power source On and Off to avoid electric shock.

4.1 Preparations for Disassembly



DANGER

- * Before undertaking disassembly of the compressor, remove the refrigerant from inside the compressor crankcase and confirm that the internal pressure is equal to the atmospheric pressure. If high pressure refrigerant gas or oil remains inside the crankcase, there is a possible of blow off during disassembly, posing a danger of explosion, suffocation, fainting, poisoning, etc.
- * If the refrigerant is an irritant or poisonous type, neutralize the refrigerant using nitrogen or dry air to reduce the density as much as possible. Care must be taken when removing covers, etc. Do not face the gas outlet port when opening, taking the risk of absorbing gas, poisoning or fainting, etc. into consideration.
- * Should a considerable amount of refrigerant be exhausted inside a closed space such as the machine room, it can lead to suffocation or poisoning. Countermeasures for ventilation and refreshment of air inside the closed room are essential.

4.1.1 Disposal of Refrigerant

If refrigerant is discharged into the atmosphere improperly, it will have a damaging effect on the environment, causing contamination, e.g., ozone layer destruction and global warming from HCFC refrigerant, foul odor and toxicity from NH_3 refrigerant, a global warming effect from CO_2 refrigerant, potential explosion and fire from combustible gases, etc. If a lot of the refrigerant is released inside the closed machine room on the other hand, there is a risk of suffocation or poisoning. Regardless of the kind of refrigerant used, proper disposal is essential for management of the plant system. Refrigerant released to the open air should be controlled at a minimal level.

(1) If there is no problem operating the refrigerating compressor.

- a) Close all valves connected to the compressor suction port side such as the suction stop valve, oil return valve from the oil separator, starting pressure equalizing valves, etc. (except for the gate valves for the pressure gauges).
- b) Make the discharge stop valve ready to close immediately.
- c) When a low pressure control (protection) mechanism is equipped, release so that the mechanism does not function even after the pressure goes down. For automatic systems, it is possible that operation is not available after short circuiting the temperature controller, etc. Confirm the system first and select manual operation.
- d) Operate the compressor so that it stops when the suction pressure reaches 0MPaG (or 0kgf/cm²G).
- e) Leave the compressor as is, so refrigerant mixed in the compressor oil can evaporate. The pressure will rise little by little as this occurs. Operate the compressor again and stop when suction pressure reaches 0MPaG (0kgf/cm²G).
- f) Repeat procedures from a) through e). When suction pressure does not go up more than 0.02MPaG (0.2kgf/cm²G), operate the compressor to reach Vacuum.
- g) When the Vacuum level advances to a point where oil pressure does not go up, stop the compressor and close the discharge stop valve.
- h) Turn off the main power supply switch. Put up notices on the breaker, control system, etc. to ensure that the power supply is never turned on or connected by mistake.

(2) When the compressor cannot be operated, carry out the following measures:

- a) Close the suction stop valve, discharge stop valve, oil check valve for returning oil from the oil separator and all other valves (except the valves for pressure gauges) to isolate the compressor from other components.
- b) Disposal of refrigerant differs according to the kind of refrigerant. If it is HCFC refrigerant, recover the refrigerant using an alternate compressor of the same type. In case of NH_3 refrigerant, recover the refrigerant as above or neutralize the refrigerant in water before disposal. In case of CO_2 refrigerant, use a recovery machine. When carrying out recovery or neutralization work, a special neutralizing agent, special knowledge and experience are required. Contact Mayekawa Mfg. Co., Ltd. or the nearest representative in your region.

4.1.2 Cautions on Disassembly



- * Always uses the regular specified hand tool kit only for compressor disassembly and assembly work. Use of inappropriate, worn or damaged tools may lead to serious injury.
- * Use the maintenance hand tools only after fully understanding utilization and familiarization with the characteristics thereof.
- * Prepare safety shoes, safety goggles, gas mask and other proper safety wear. Use these as necessary.
- * Spare parts are covered with oil to prevent rusting and are prone to slipping when handled. Care must be taken when handling spare parts to avoid physical injury.
- * Ensure sufficient spare parts to carry out the maintenance work. Care must be taken not to place spare parts or wiring cable carelessly on the floor as this poses a hazard.
- * Always keep the floor clean. If the floor is stained with oil, etc., it poses a risk of causing falls. Wipe up all stains and foreign matter immediately.
- * Keep hand tools clean during the service work. Dirty tools can leave harmful residue in the compressor, leading to serious malfunction and damage during operation.

- a. Care must be taken when removing component parts to avoid scarring, bending or other damage.
- b. After removal, place spare parts on a clean worktable in order.
- c. When washing spare parts, use light oil or fine washing oil. Compressor lubricating oil can also utilized as a detergent by warming to around 40~50°C. (Special care must be taken to prevent fire)
- d. After washing spare parts, dry completely using the air pressure, a sponge or clean white cloth, then coat with new lubricating oil. Do not use cloth that easily sheds fibers.
- e. When the cylinder assembly is disassembled, place all parts for each cylinder in order. Special care must be taken with parts of the connecting rods. Do not mix with others as the combination of connecting rod and rod cap differ and are not interchangeable.
- f. Locking bolts should be placed in order.

4.2 Compressor Oil Discharge

After refrigerant gas recovery from the compressor is completed, lubricating oil inside the compressor should be drained. Connect a pipe or hose to the oil supply/drain valve located on the right side below the oil pump and drain the oil into a container of sufficient size. Nitrogen gas or dried air can be used to raise pressure to around 0.1MPa to facilitate removing oil. Do not forget to remove the gas and return pressure inside the compressor to atmospheric pressure.



Fig.4.1 Oil supply/Drain Valve

4.3 Disassembly of Rotating Portions

Remove the coupling spacer and pull off the coupling flanges (hub) using a pulley remover. Both the crankshaft and the coupling hubs are tapered and can be removed easily.

4.4 Head Cover (49 & 50)

4.4.1 Removing Jacket Cover (53)

※Only for water-cooled type head cover (In case of air-cooled type, proceed to 4.4.2)

- a. In case of water-cooled head cover, remove the jacket hose first.
Before removing the head cover confirm and record the number, place and the mounting direction of the head cover. If the place or the direction of the head cover is changed on disassembly, reassembly of head cover and mounting of the water hose will become difficult.
- b. Remove all bolts, and take off the jacket cover.
※ When taking off the jacket cover, water remaining inside the jacket may spill out.

4.4.2 Removing Head Cover (50 & 49)

Net weight of Head Cover: Approx. 15kg.

- a. Remove the head cover bolts from top to bottom in order to prevent scarring and damage of the cylinder and other components if dropped on the floor and to prevent foreign matter entry into particles into the crankcase.
- b. Remove the upper bolts first, then mount a safety bolt.
※ The safety bolt can easily be prepared using a longer one than the locking bolt by cutting down the top of the bolt a little. A fully threaded bolt can also be used as a safety bolt.
- c. Loosen the remaining bolts and remove all bolts so that the head cover pushed up slightly from the crankcase.
- d. The head springs (117) are fixed into grooves in the discharge valve cages (109), but it is possible for them to slip out from the groove and fall. Remove the head cover while taking care to prevent the head springs slipping out of the grooves.
- e. Finally remove the head springs.



Fig. 4.2 Head Cover, Water-cooled type (50)

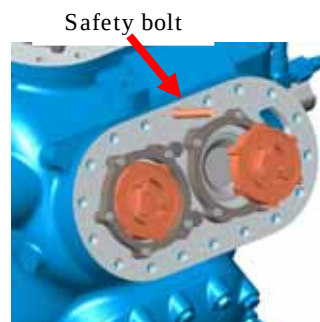


Fig.4.3 Head Cover, Air-cooled type (49)

4.5 Discharge Valve Assembly and Valve Plate (110)

4.5.1 Discharge Valve Cage Assembly (i.e., D.V. Cage)

- a. The discharge valve (plate valve) is fitted in the discharge valve cage assembly and positioned by the discharge valve caged guide (74).
- b. Raise the discharge valve cage upward in parallel with the discharge valve caged guide. It is difficult to remove it diagonally.
- c. Even when the discharge valve cage guide is not removed, disassembly work is still



- possible. If it is necessary to remove the discharge valve cage guide, pull off along with the positioning pin (75-1).
- The top of the valve plate (110) functions as an external valve seat for the discharge valve after the discharge valve cage is removed. Care must be taken to keep the surface free of damage.
 - The discharge valve seat (111) is mounted on the discharge valve cage using three locking bolts (112). Bend back the claws of the locking washers, then remove the locking bolts. The discharge valve seat is positioned using a parallel pin. The discharge valve seat (111) may be stuck to the discharge valve cage (109) but can be removed by turning a little. When the discharge valve seat is removed, the discharge valve (Plate Valve) comes out. The discharge valve springs (116) are mounted in the spring holes of the discharge valve cage (109). Remove the discharge valve springs by pulling and rotating.

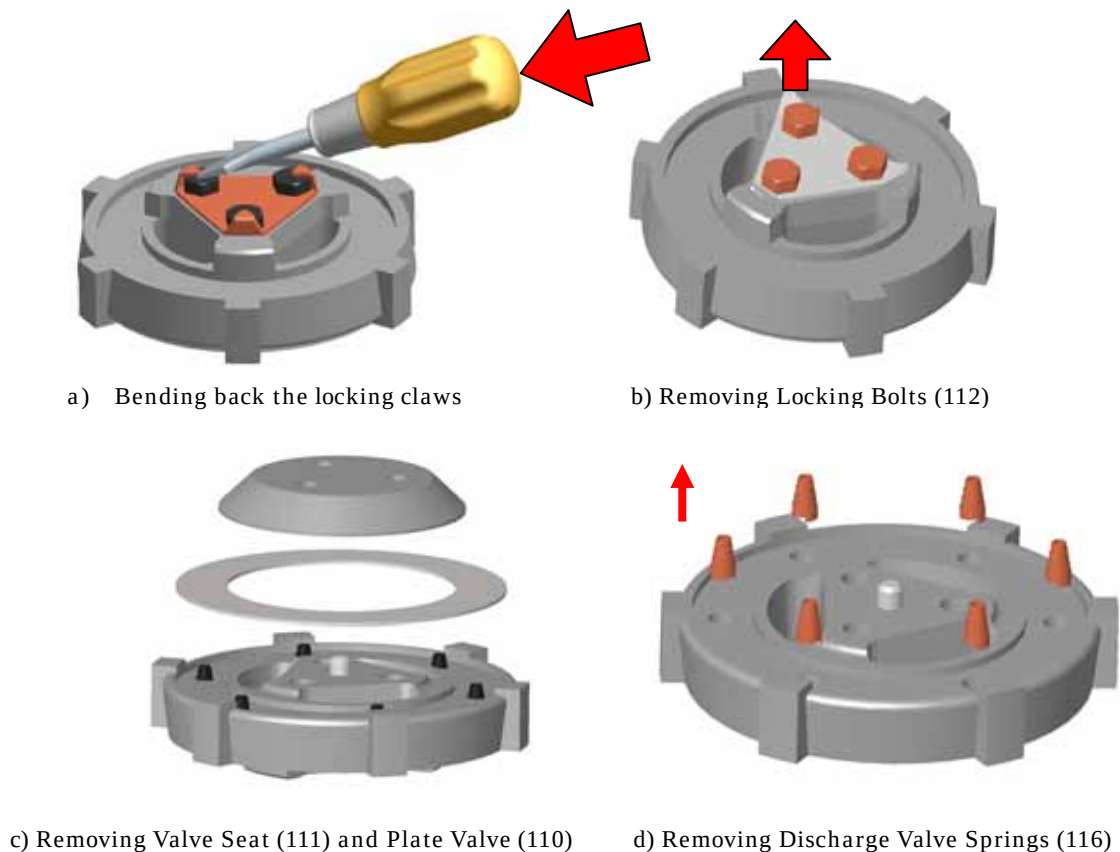


Fig.4.5 Disassembly of Discharge Valve Assembly

4.5.2 Valve Plate (73)

Net Weight: Approx. 6kg.
(Together with the cage guide)

- The valve plate is fixed to the crankcase with head cover locking bolts(52) and two positioning pins (73-1) as shown in Fig.4.6.
- Pull up the valve plate perpendicularly against the mounting surface.



Fig.4.6 Removing Valve Plate (73)

4.6 Hand Hole Cover (45)

Net weight: Approx. 6kg.

- Remove the upper bolt first and replace with a safety bolt.



- b. Remove all remaining bolts. When removing the last bolt, care must be taken while supporting the cover by hand to prevent it falling on the floor. When removing the hand hole cover from the crankcase, oil remaining inside will spill out. Place an oil drain pan below the hand hole cover to catch the oil. If the hand hole cover gasket sticks to the crankcase, tap the cover with a plastic or copper hammer or mallet.
Fig.4.7 Removing Hand Hole Cover (45)
- c. The oil sight glass(164) is mounted on the hand hole cover using three bolts (167). Remove these three bolts, and pull off the sight glass.

4.7 Cylinder Assembly

(Cylinder Sleeve, Piston, Connecting Rod, etc.)

Net weight of assembly: Approx. 4 kg.

4.7.1 Drawing out Cylinder Assembly

- a. The width of the connecting rod big end is larger than the diameter of the cylinder sleeve (61). The connecting rod (77) should, therefore, be removed from the crankcase together with the cylinder sleeve and the piston (85) as an assembly.
- b. First turn the crankshaft until the piston is positioned at bottom dead center. When turning the crankshaft, the cylinder mated with the piston goes up together and it is possible for it to slip out from the crankcase. Hold down the cylinder sleeve when turning the crankshaft. The remaining cylinders can be secured from lifting up by temporarily locking the valve plate.
- c. Remove the No.2 and No.1 nuts (80) of the connecting rod bolts (78) in order, then remove the big end rod cap. Care must be taken not to lose the washer (79) and nuts (80).
- d. Push the cylinder up by hand from the hand hole of the crankcase, so the cylinder comes free of the crankcase. Hold the edge of the cylinder sleeve by a hand and pull out the assembly being careful to not let the connecting rod touch the crankcase.
- e. Remove all remaining cylinder assemblies following the same procedure. All cylinder assemblies removed from the crankcase should be stored in correct order for reassembly. All caps, bolts and nuts and washers should be grouped with their respective cylinder assemblies.



Fig.4.8 Removing lock nut



Fig.4.9 Removing Rod Cap



Fig.4.10 Removing Cylinder Assembly

4.7.2 Piston (85) and Connecting Rod (77) Removal

- a. After removing the bolts from the big end, place the cylinder assembly on a work table cushioned by a rubber sheet or clean cloth with the cylinder seat (flared end) facing downward. The connecting rod retainer bolts (78) should be drawn out before this is done.
- b. Holding the cylinder sleeve with one hand, draw out the piston while supporting the connecting rod big end with the other hand. When the cylinder assembly is resting on an air-tight surface such as a rubber sheet, it may be difficult to draw out the piston because of internal vacuum. Turn the assembly slightly to let in air to facilitate removal.

4.7.3 Cylinder Sleeve (61)

- a. It is usually unnecessary to disassemble the cylinder sleeve unless the sleeve itself or the cam ring (62 & 63) must be replaced. If the retaining ring (65) fitted in the lower groove on the sleeve is removed, the cam ring will drop off.
- b. Two different cam rings are used, one left downward slanted and the other right downward slanted. The mounting positions of these cam rings are fixed, so care must be taken to select the mounting position, or else the unloader mechanism may not function.
- c. Remove the unloader lift pin stop ring (E-ring) (70) from the lift pin (68) and remove the lift pins from the cylinder sleeve. Be careful not to lose the springs (69) and E-rings (70).



a) Removing Retaining Ring (65)



(65)

b) Removing Cam Ring (62L & 63R)

Retaining ring



c) Removing E-Ring (70)



d) Removing Lift Pins (68)

Fig. 4.11 Disassembly of Unloader Mechanism

4.7.4 Piston (85) and Connecting Rod (77)

- Position the piston with the crown facing down and the rod end facing up. Remove the piston pin lock spring (87) using a pair of snap ring pliers.
- Holding the piston and connecting rod securely, push the piston pin (86) out to separate the piston and connecting rod.
- Do not remove the connecting rod large end bearing halves (84) unless replacement is necessary. The small end of the connecting rod is furnished with a wrapped bearing.
- Each set of connecting rod large end and cap is engraved with mating marks. Be sure to keep the rod and cap pairs together.



Fig. 4.12 Removing Piston Pin (86)

4.7.5 Piston Rings (89 & 90)

- Do not remove the piston rings unless replacement is necessary.
- Utilize hooks or a pair of piston ring pliers or bindings to expand the ring end gap on the same plane to facilitate removal of the ring. (Refer to Fig.4.13)
- Torsional expansion of the ring may lead to deformation of the ring, resulting in oil leakage after reassembly or breakage due to excessive deformation.



4.8 Unloader Mechanism Disassembly

- Remove the coil portion of the unloader solenoid valve (remove the nut which secures the coil) and take off the unloader cover bolts. Since the cover will move outward due to the pressure of the unloader spring, be sure to keep a good grip on the cover.
- Remove the unloader cover so that the unloader piston, push rod and spring can easily be removed. The length of each push rod differs according to the number of the cylinder it is mounted on. Keep a record of the cylinder number and the unloader cover number as well as the direction.
- The locking ring and the washers of the unloader push rods should never be removed but left in place.



Fig.4.14 Removing Solenoid Valve Coil



Fig

1

4.9 Mechanical Shaft Seal (32) Disassembly

4.9.1 Removing Shaft Seal Cover (26)

- Before removing the shaft seal cover, place an oil pan under the cover to catch any oil remaining inside.
- Loosen all shaft seal cover locking bolts (28). If the gasket (27) is not stuck to the crankcase, a small opening is created between the seal cover and the casing by the spring power of the shaft seal. Except for two opposite locking bolts, remove all remaining bolts, then remove the remaining two bolts evenly while securing the seal cover.
- In case the shaft seal cover is locked to the crankcase because the gasket is stuck to the casing, turn the cover by hands left and right while loosening the bolts. If the cover does not move at all, insert a minus driver into the crevice and pry off evenly.
- Draw out the shaft seal cover along to the crankshaft axis.
- The shaft seal cover is furnished with a floating seat made of carbon together with the O-ring. The floating seat can easily be removed by inserting a finger between the seat and the seal cover, but care must be taken because the seat can easily be broken.



Fig. 4.16

4.9.2 Removing

- Remove the shaft seal cover (26) to expose the mechanical seal assembly (32). The mechanical



shaft seal assembly is connected to the shaft by a spring retainer positioned by a knock pin. It consists of a seal ring, O-ring, drive collar and seal spring.

- b. Grasp the shaft seal assembly with both hands and draw out while securing the seal collar positioned inside.
- c. Dismantle the seal assembly, arranging the parts in order and store with care. The seal ring frictional face should be protected to prevent damage.

Fig. 4.18 Shaft Seal removal

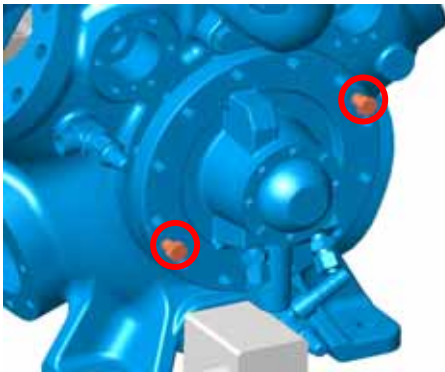
4.10 Bearing Housing (8)

4.10.1 Removing Bearing Housing (8)

Net weight: Approx. 15kg

※ The clearance of the crankshaft in the thrust direction should be measured before removing the bearing housing (8). Push the shaft in the crankcase close to the seal side or the pump side and measure the clearance with a thickness gauge inserted between the opposite shaft end face and the metal. Alternatively, push the shaft to the pump side and measure the distance moved in the thrust direction using a dial gauge mounted on the end of the crankshaft.

- a. Remove all bearing housing retainer bolts (10-1 & 10-2). Two tap holes are prepared on the flange portion of the bearing housing (refer to Fig. 4.19-a) below). Insert and screw two retainer bolts (10-1) into the tap holes to press off the bearing housing.
- b. After pushing the bearing housing off somewhat with the tap hole bolts, grasp the bearing housing with both hands and pull off. The shaft fitting for the bearing inside the bearing housing may sometimes come out together with the housing. Secure the crankshaft by hand through the hand hole or from the seal side to prevent the crankshaft being pulled out together with the bearing housing.



a) Resetting two bolts into the tap holes



b) Removing bearing housing

Fig. 4.19 Removing the Bearing Housing

4.10.2 Pump side Main Bushing (12) and Thrust Washers (29-1 & 29-2)

- a. The main bushing is press fitted into the hole in the main bearing housing. In appearance it resembles a thinly coiled bushing. It is usually unnecessary to remove the bearing housing assembly unless replacement is required.
- b. The thrust washer (29-1 & 29-2) is a circular-shaped metal plate with a backing

sheet. It is fitted to the thrust face of the bearing housing with a lock pin.
 c. When replacing the thrust washer, push out the metal with a fine driver.

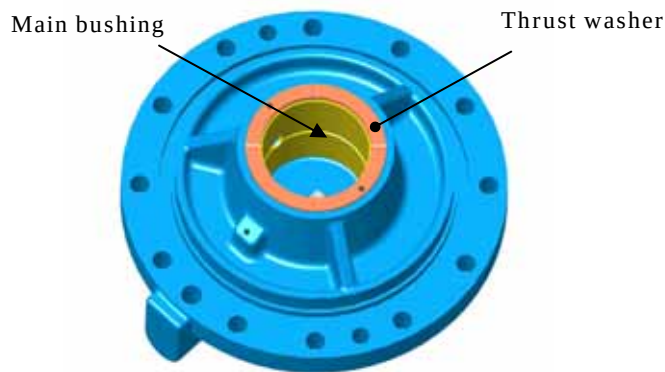


Fig. 4.20 Main Bushing (12) and Thrust Washer (29)

4.10.3 Oil Pump

The oil pump assembly is mounted on the bearing housing (8). If the pump shaft does not rotate smoothly or is experiencing a drop in performance, replace the complete bearing housing assembly.

4.11 Crankshaft (2)

Net weight : Approx. 30 kg

- a. After removing the bearing housing (8), the crankshaft is held in place only by the main bushing (12) on the shaft seal side. With an assistant to support one end of the crankshaft, lift and draw out the shaft until the bearing housing side balance weight can be rested on the bearing housing mounting flange.
- b. Next move to position the center arm of the crankshaft on the bearing housing mounting flange as shown below. Finally, remove completely.
- c. After the crankshaft has been removed, place on a cloth or rubber sheet to prevent damage to the bearing journals and the mechanical seal mounting portion.



Fig. 4.21 Removing the Crankshaft (2)

4.12 Seal side Main Bushing (12) and Thrust Washer (29)

The seal side main bushing (main bearing) is press fitted into the crankcase. In case of replacement, utilize the specified tools only.

The seal side thrust washer is also fixed to the cavity of the crankcase using the same locking pins (13) as for the bearing housing side washer, so remove it.

4.13 Oil Filter (119)

The oil filter is mounted inside the crankcase with a hexagon head nipple (118). Remove the oil filter together with the nipple. Since the wire material of the filter element is rather fragile, do not apply an excessive force when removing it.

4.14 Suction Filter (154)

The suction filter is mounted inside the suction elbow (175). The suction filter should be cleaned during periodic inspection. Special care must be taken to confirm that there is no breakage of the filter or the joint portion. Remove the bolts of the piping side flange and the locking bolts for the crankcase and remove the suction elbow (Approx. 7kg), then remove the suction filter from the elbow.



Fig. 4.22 Removing Oil Filter (119)



Fig. 4.23 Removing Suction Filter (154)

4.15 Oil Pressure Regulating Valve (22)

The oil pressure regulating valve does not normally need to be disassembled. If removed or replaced with a new one, carry out adjustment of the regulating valve.

4.16 Pressure Equalizing Pipe (305)

There is a pressure equalizing hole between the suction chamber and the crankshaft chamber of the compressor to prevent any difference in pressure. The pressure equalizing pipe is mounted in the pressure equalizing hole to prevent oil inside the crankshaft chamber from entering the suction chamber. A T-shaped squared pipe is mounted on the crankshaft chamber side, and a regular cylindrical pipe is mounted on the suction chamber side. The cylindrical pipe end is furnished with a welded nut for tightening. The pressure equalizing pipe need not be disassembled, but confirm that it is tight.

4.17 External Oil Filter (-)

Mark the oil flow direction at the top of the flange to prevent error during reassembly. Remove the tightening bolt (Cap Bolt) located on the upper portion and pull off the flange. Inside the flange you may find an accumulation of foreign matter together with compressor lubricating oil. Open the drain plug at the bottom to drain the oil and foreign matter. Thereafter clean the inside using kerosene. The filter element can be easily removed by pulling up on the mounting portion of the filter. If dirt or deformation is found, replace the filter element with a new one.

5. Reassembly

- ※ All parts disassembled from the compressor should be repaired or replaced according to the inspection and replacement standards given later in this manual. Reassembly work is essentially carried out in the reverse order of disassembly. When ordering spare and

replacement parts, specify the number of the part indicated in the exploded view and parts list, the name of the part, the model and the serial number of the compressor.

5.1 Cautions on Reassembly



DANGER

- * Do not use oxygen or combustible gas for air-tight testing as there is a risk of explosion.



CAUTION

- * Always use the normal specific hand tool kit only for disassembly and assembly work of the compressor. Use of inappropriate, worn or damaged tools may lead to serious injury.
- * Use the maintenance hand tools only after fully understanding and familiarization with the characteristics thereof.
- * Prepare safety shoes, safety goggles, gas mask and other proper safety wear. Use these as necessary.
- * Spare parts are covered with oil to prevent rusting and are prone to slipping when handling. Care must be taken when handling spare parts to avoid physical injury.
- * Ensure sufficient spare parts to carry out the maintenance work. Care must be taken not to place spare parts or the wiring cable carelessly on the floor as this poses a hazard.
- * Always keep the floor clean. If the floor is stained with oil, etc., it poses a risk of causing falls. Wipe up all stains and foreign matter immediately.
- * Keep the hand tools clean during the service work. Dirty tools can leave harmful residue in the compressor, leading to serious malfunction and damage during operation.
- * Do not use CO₂ gas for air-tight testing of an ammonia (NH₃) compressor as there is the possibility of generation of ammonia carbonate, which is harmful to the system.

- When replacing a part, confirm that the new part is in fact the correct one before starting reassembly.
- Clean all parts in kerosene and remove any visible rust and degraded oil with fine emery paper. With the crankshaft, remove the locking plug and clean inside the oil passage. Remount the locking plug immediately after cleaning to avoid loss. (Cleaning of the seal tape is not necessary).
- Prepare an oiler containing fresh lubricating oil and apply oil to all friction surfaces just before carrying out reassembly.
- Dry all washed parts with compressed air or wipe dry using a cotton or non-woven service sheet. Do not use cloth materials such as wool or synthetic fiber as these do not absorb oil well and may leave fibers on the parts, causing later problems. Care must be taken to ensure that the oil passages in the parts are not closed by applying air or using a viewing light.
- The gaskets used between mating surfaces should be coated with lubricating oil on both sides before installation in order to assure airtight sealing. When replacing a gasket, scrape all old gasket residue off of the mating surfaces before mounting the new gasket.
- All hand tools used during disassembly should be cleaned before commencing reassembly work.
- When mounting a cover, care must be taken to ensure that the gasket is properly positioned. Be careful to avoid blocking oil ports when mounting gaskets.
- Fasten all bolts lightly then retighten to the specified torque in a crisscross pattern.

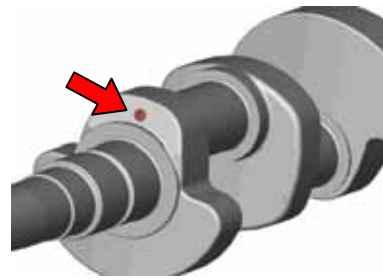


Fig.5.1 Locking Plug of Crankshaft.

5.2 Crankcase Reassembly

- Check the main bushing (12) and install the thrust washers (29), then supply lubricating oil to all portions.
- When the main bushing is to be renewed with a new one, do not drive in, but press-fit using the special main bushing specific press-fitting tool. After press-fitting the main bushing, confirm that the oil hole of the bushing and the oil hole on the crankcase are aligned. The oil hole in the crankcase is located at the right oil pressure regulating valve observed from the shaft seal side. Even though the two oil holes do not mate perfectly, no functional problem will occur. If the two holes do not mated mostly or completely, draw out and press-fit once again.
- Mount the oil filter while turning the hexagonal head nipple. Do not push on the filter element, otherwise it may be deformed.

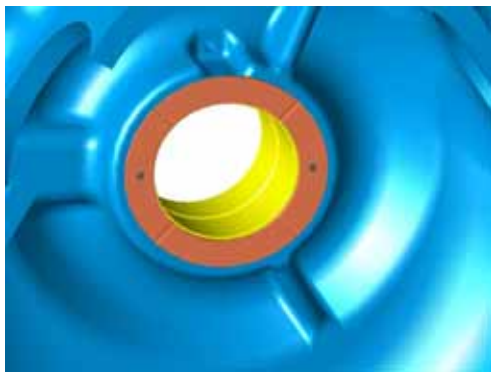


Fig. 5.2 Mounting Thrust Washer (29)

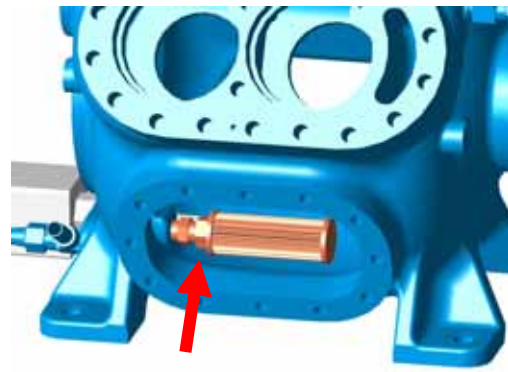


Fig. 5.3 Mounting Oil Filter (119)

5.3 Crankshaft (2) Reassembly

- Apply a generous coating of lubricating oil to the crankcase journals. Place the crankshaft balance weight on the flange portion of the bearing housing.
- Insert the crankshaft into the crankcase until the center arm rests on the bearing housing flange. When carrying out this work, be careful not to damage the main bushing with the shaft end as it enters the main bushing.
- When the balance weight of the crankshaft on the oil pump side reaches the flange portion, the crankshaft bushing portion is ready to accept the main bushing. The crankshaft end is visible at the shaft seal side. Insert the shaft into the main bushing while supporting both ends of the crankshaft. Move the crankshaft parallel to the shaft core.
- Push on the shaft in until it contacts the thrust face.
- Locate the groove for the oil pump in a vertical position by rotating the crankshaft.



Fig. 5.4 Reassembly of Crankshaft

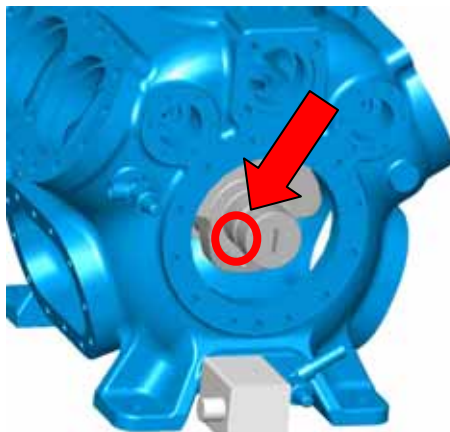


Fig.5.5 Groove portion of Crankshaft

5.4 Bearing Housing (8) Reassembly

- If the main bushing and oil pump must be replaced, complete renewal of the bearing housing assembly is generally carried out.
- Fit the thrust washer (29) on the bearing housing (8) mate with the thrust washer locking pin (13). (Refer Fig.5.6)
- Position the oil pump vertically to mate with the crankshaft end.

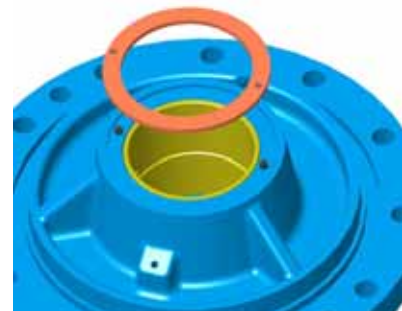


Fig.5.6 Mounting Thrust Washer

- Apply a generous coating of lubricating oil to the journal portion of the shaft and

- the oil supply hole in the crankshaft.
- Mount the bearing housing gasket (9) on the surface of the housing (8). Be careful to avoid blocking the oil hole.
 - Mount the main bushing (12) on the crankshaft (2) and then, raising the bearing housing slightly, push the crankshaft into the crankcase.
 - The pump shaft fits into the crankshaft groove during the final 20mm of travel. If the pump shaft does not fit in the crankshaft groove, the bearing housing will stop movement and it will become difficult to push in the shaft. Turn the bearing housing left and right slightly to ensure a tight fit. (If the bearing housing is turned while pushing down, the pump shaft is also rotated, therefore turn the bearing housing while pressing on the pump shaft or turn the bearing housing after pulling on the bearing housing slightly.)
 - When the shaft fits into the groove properly, it can easily be pushed into the crankcase. Fasten the bolts lightly, confirm that the oil pump is a good fit while rotating the crankshaft by hand, then tighten the bolts to the specified torque.
 - Move the shaft in the axial direction and measure the thrust gap.

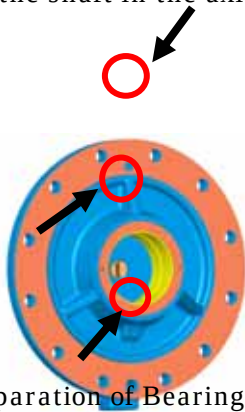


Fig.5.7 Preparation of Bearing Housing



Fig.5.8 R

ing

5.5 Mechanical Shaft Seal (32) Reassembly

- Fit the floating seat on the shaft seal cover (26). Apply oil to the O-ring and press evenly onto the seat.

[Caution] The floating seat, made of carbon, is weak against impact load. Care must be taken not to strike the floating seat with a solid tool or to allow the friction surface to become scratched.



Fig.5.9 Reassembly of Floating Seat

- Position the shaft seal collar drive pin of the crankshaft upward and position the seal collar notch of the shaft seal assembly to mate with the locking pin, then push in the assembly. Confirm that the locking pin engages with the notch.
- Apply oil to the gasket and stick it to the seal cover, being sure to mate the oil holes.
- Apply oil all over the surface of the shaft seal assembly and the floating seat, then mount the shaft seal assembly in the shaft seal cover (26) while maintaining perpendicular orientation with respect to the shaft. Special care must be taken to ensure that the shaft end and the friction surface of the floating seat do not touch each other.
- While securing the shaft seal cover by hand, install the shaft seal cover on the crankshaft with bolts (28) and tighten to the specified torque. Turn the crankshaft by hand to confirm that the shaft seal cover is correctly positioned. When the crankshaft cannot be rotated or feels too heavy to rotate, there is possible that the shaft seal is not mounted correctly. Repeat disassembly and reassembly work.

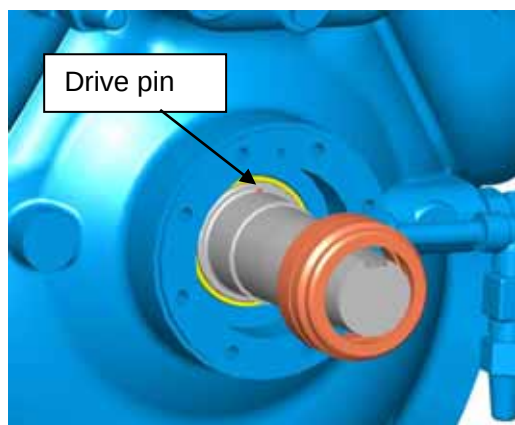


Fig.5.10 Reassembly of Shaft Seal Assembly



5.11 Reassembly of Shaft Seal Cover

5.6 Reassembly of Cylinder Assembly

5.6.1 Piston (85), Piston Rings (89 & 90) and Oil Ring (100)

a. Mounting the Piston Rings (89 & 90) and Oil Ring (100) on the Piston (85)

Table 5.1 Composition of Piston Rings and Oil Ring

		CO ₂ /HFC System	Ammonia
1 st. (top) ring	Piston ring	FC-P (Black)	GA-P (Yellow)
2 nd (middle) ring	Piston ring	FC-P	
3 rd (bottom) ring	Oil ring	FC-CO	

The piston rings should be mounted starting with the oil ring (bottom) and moving upward in order.

- b. When installing the piston rings on the pistons, care should be taken to avoid excessively widening or distorting the rings. After installing each ring into its respective ring groove, confirm that the ring moves smoothly in the groove.
- c. The ring end gaps of the three rings should be spaced 120~180 ° apart.



a) Mounting Oil Ring (100)



b) Mounting Piston Rings (89 & 90)

Fig.5.12 Mounting Piston Rings and Oil Ring.

5.6.2 Piston (85) and Connecting Rod (77)

- Place the piston upside down and attach the connecting rod small end using the piston pin (86), as shown in Fig.5.13. Apply oil to the bearing halves of the connecting rod big end, the bearing portion of the connecting rod small end and the piston pin (86). The piston pin can easily be mounted in the pin hole of the piston.
- Mount the piston pin lock spring (87). Confirm that the lock spring is exactly fixed into the spring groove of the piston.
- The bearing halves are designed to use in common suitable to use for upward and downward movements. The bearing half metal locking claw should mate with the notch of the connecting rod cap (see below).



Fig. 5.13 Reassembly of Piston and Connecting Rod Assembly

5.6.3 Cylinder Sleeve (61)

- If the cylinder sleeve is replaced, first drive the positioning pin (61-1) into the flared portion of the cylinder. (The positioning pin is also called the “Retaining pin.”)
- Mount the unloader lift pins (68), the lift pin springs (69) and finally install the lift pin E-rings (70) in order.
- Mount the cam rings (62 & 63) and secure with the retaining ring (65). Care must be taken to ensure that the slant direction of the cam ring is correct. Turn the cam ring to engage the notch on the cam ring with the retaining pin (61-1). The most suitable position of the notch on the cam ring is placed on the slope of the cam ring. Regarding the relationship between the mounting position of the cam ring and the slant as well as the position of the retaining pin (61-1), refer to the Table 5.2 and the figures below.

- When the slope of the cam ring is right downward slanted:

The notch on the cam ring should be located at 45 ° turning leftward from the positioning pin.

- When the slope of the cam ring is left downward slanted:

The notch on the cam ring should be located at 45 ° turning rightward from the positioning pin.

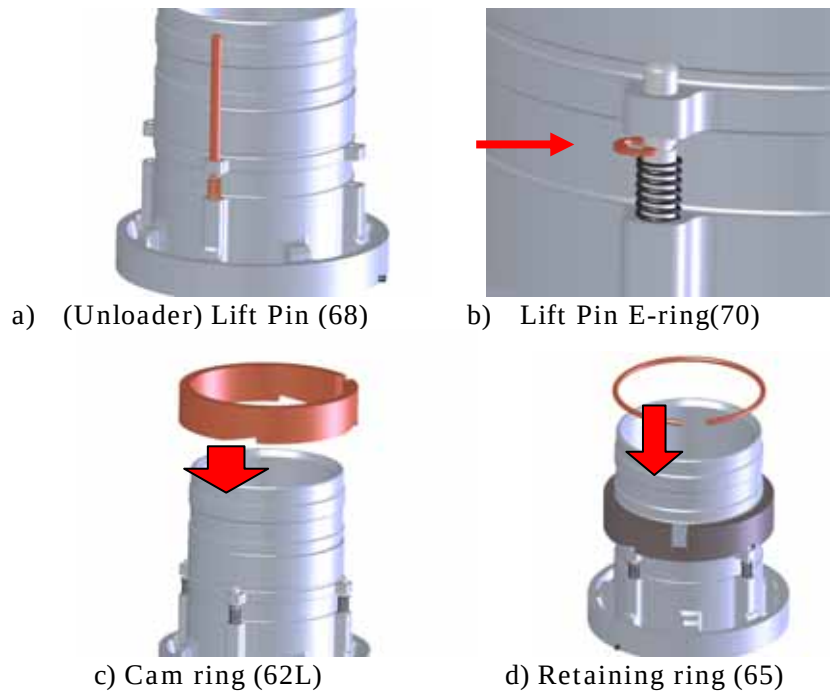
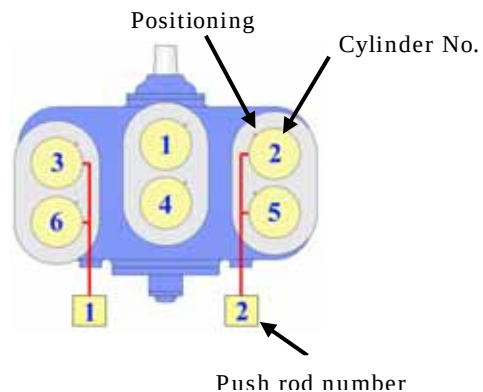


Fig.5.14 Unloader mechanism reassembly

Model 6HK Compressor

Table 5.2 Cam ring remounting

Piston No.	6HK
1	—
2	Right downward slanted
3	Left downward slanted
4	—
5	Right downward slanted
6	Left downward slanted



5.15 Cylinder Number

5.6.4 Cylinder Sleeve (61) and Piston/Connecting Rod Reassembly

a. Place the cylinder sleeve with the seat side facing down. If the reassembly work for the cylinder sleeve, piston and connecting rod is carried out on a rubber sheet or fine metal surface, air inside the assembly may be difficult to remove when the piston is pushed down. In such case, turn the cam ring to raise the lift pins (68).

b. Apply lubricating oil to the piston and piston rings while holding the connecting rod portion of the piston/ connecting rod assembly, then insert into the cylinder sleeve approximately half-way. The piston ring section of the piston is inserted into the cylinder from the tapered end of the cylinder sleeve while squeezing the rings.



Fig.5.16 Reassembly of piston/ connecting rod assy.

- c. Stop insertion of the piston midway into the cylinder sleeve so the cam ring can be positioned in the same location as before disassembly
- d. Place the cylinder assembly, connecting rod cap, bolts and nuts in lots for each different cylinder to avoid mixing them

5.7 Unloader Mechanism Reassembly

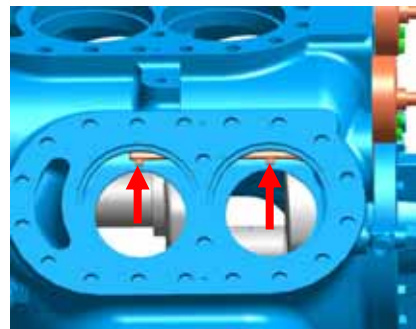
- a. Mount the washers (143) and the springs (142) on the unloader push rods (135 & 136), then insert into the crankcase.
- b. Mount the unloader piston (145) and secure the piston with the unloader cover (146). The unloader cover should be eccentrically slipped out somewhat as showed in figure. When the piston end surface and the surface of the crankcase are even, the pin portion of the unloader push rod is placed at the center of the cylinder, thus positioning is achieved. Should the unloader cover be mounted normally without slipping out, the pin portion of the unloader push rod is slightly slipped out from the center portion to the direction of the unloader cover. If a different length of push rod is selected, the pin portion of the push rod will not reach the center portion of the cylinder, resulting in unloader mechanism malfunction.



Fig.5.17 Mounting Unloader Mechanism



a) Mounting Unloader Cover (146)



b) Position of Unloader Push Rod (135 & 136)

Fig.5.18 Positioning of Unloader Push Rod (135 & 136)

5.8 Mounting Cylinder Assembly in Crankcase (1)

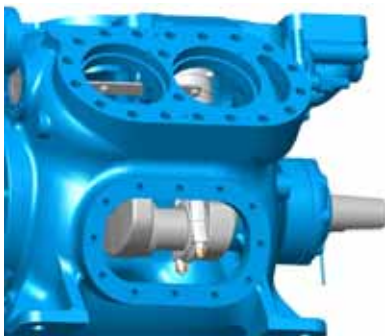
- a. The cylinder assembly should be mounted in the crankcase in order of cylinder number (ref. to Fig. 5.15). Alternatively, mount the cylinder assemblies from the end of the crank pin portion in order (refer to 3.2 Crankshaft and figure on Page 7). Confirm the slant direction of the cam ring, the position of the notch on the cam ring and the direction of the retaining ring once again (refer to 5.6.3, Cylinder Sleeve). Install the retainer bolts (78) on the connecting rod (77).
- b. Position the crank pin end of the cylinder at bottom dead center, then apply lubricating oil to the crank pin and the bearing halves. While maintaining the piston at the top of the cylinder, insert the cylinder assembly into the cylinder hole in the crankcase. When the notch of the cam ring is mated with the pin of the unloader push rod, fix the retaining ring of the cylinder sleeve with the groove on the crankcase, then push the cylinder assembly into the crankcase.

- c. While securing the connecting rod by hand, push the top of the piston down to insert the piston/ connecting rod assembly into the crankcase and seat the bearing metal portion of the connecting rod big end on the crank pin portion of the crankshaft. Care must be taken not to scratch the crankshaft with the bolts. If the connecting rod rotates or slips, the connecting rod cannot be mounted satisfactorily. Move the connecting rod back and forth or right and left slightly by hand to confirm that the rod is properly fitted.

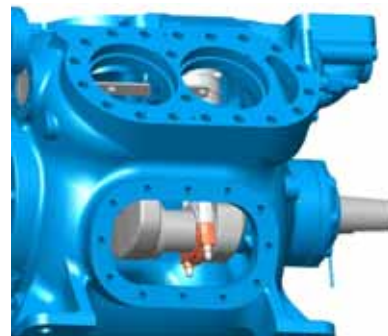


Fig.5.19 Mounting of Cylinder Assembly

- d. Confirm the rod cap fitting direction, mount, and install the washers (79) and the first nut (80) after confirming that the mating numbers on the big end and the rod cap match. Tighten the bolts alternately to the specified torque.
- e. Fit the second nut on each bolt and tighten to the specified torque.
- f. Rotate the crankshaft once after installing each assembly to confirm smooth movement. At this time hold the cylinder by hand, otherwise, not yet being secured, it will move somewhat with the piston due to the frictional force of the piston rings. In this case, it is possible to retain the cylinder sleeve temporarily using a washer and bolt (73-1) for the valve plate (73).



a) Mounting connecting rod cap on crankshaft and tightening No.1 nut



b) Tightening No.2 Nut

Fig.5.20 Tightening Connecting Rod Nuts

5.9 Valve Plate (73) Reassembly

- a. Mount the suction valve springs (72) in the spring holes on the suction valve side of the valve plate (73). As the outer diameter of the spring is somewhat larger than the spring hole, it must be twisted into the hole.
- b. Mount the suction plate valve (71) on the valve plate (73) and secure the suction plate valve using the valve clip (valve gland) contained in the hand tool kit. Suction valve plate mounting should be carried out for two cylinders at the same time.
- c. Apply lubricating oil liberally on the valve plate gasket (73-4) and mount the gasket on the crankcase. At this time, confirm that the valve plate positioning pin (75-1) is set satisfactorily.

- d. Confirm that the unloader lift pins (68) are lower than the seat surface of the cylinder sleeve. When the place of the unloader cover and the unloader piston is remain unchanged after the unloader sleeve is mounted, the lift pin position will automatically be located at the lower position.



Fig.5.21 Mounting Valve Spring



Fig. 5.22 Valve clip setting



Fig.5.23 Mounting Valve Plate (73)

- e. Mount the valve plate (73) carefully making use of the positioning pin (73-3). Tighten the two center bolts (73-1), then remove the valve clip. Confirm that the plate valve (71) is positioned correctly. If the plate valve slips from the proper position, a gap is generated between the valve plate and the gasket or between the gasket and the crankcase. Confirm that there is no gap between these parts. When a cylinder is complete with an unloader mechanism, movement of the unloader mechanism can be confirmed easily because the plate valve will move up and down when the unloader piston is pushed (when the unloader piston is pushed, the plate valve closes).

[Caution] Do not open the unloader cover before mounting the valve plate, otherwise the pin portion of the unloader push rod will slip out of the notched portion of the cam ring, and be difficult to reinsert.

5.10 Remounting Unloader Cover (146)

- a. Push the unloader piston (145) by hand to confirm whether the unloader device works satisfactorily or not.
- b. Confirm that the oil passageway of the unloader cover gasket, the oil hole in the crankcase and the oil passageway of the unloader cover are properly aligned. (Also confirm that the small oil supply hole in the unloader cover is located at the top and the large oil drain hole is at the bottom.)



Fig.5.2 Mounting Unloader Cover

- c. Fix the unloader cover bolts (149) temporarily, then tighten the bolts to the specified torque in a crisscross pattern.
- d. Mount the solenoid valve coil using the locking bolt.

5.11 Discharge Valve Cage (109) Reassembly

- a. Mount the discharge valve spring (116) on the discharge valve cage (109). Twist the spring into the hole as with the discharge plate valve (110).
- b. Position the discharge plate valve (110) on the discharge valve cage (109) and secure with the discharge valve seat (111). The discharge valve seat should be aligned with the discharge valve seat positioning pin (115) located at the center.
- c. Mount the triangular discharge valve retainer (112-2) while press-fitting the valve seat (111) so the plate valve is in the correct position, then tighten the discharge valve cage (109) from the top using the bolts (112). If the bolts are tightened when the plate valve has slipped out, the plate valve will be deformed. Special care must, therefore, be taken during reassembly.
(If the end of the discharge valve retainer (112-2) is bent slightly before mounting, this work can easily be carried out).
- d. Tighten the bolts (112) to the specified torque, then confirm that the discharge plate valve (110) moves smoothly. Also confirm that the bolt heads are approximately level with the valve seat face. If they are significantly lower than the seat face with some threads exposed, the wrong bolts may have been used but if the bolt heads protrude from the seat face, they may interfere with the piston, resulting in serious damage.
- e. Bend the discharge valve retainer washers (112-2) using a screw driver, as shown in Fig.5.25-d) below. If the discharge valve cage guide (74) was mounted previously on the valve plate (73), remounting work can easily be performed.
- f. The valve plate (73) should be mounted along with the cage guide (74).

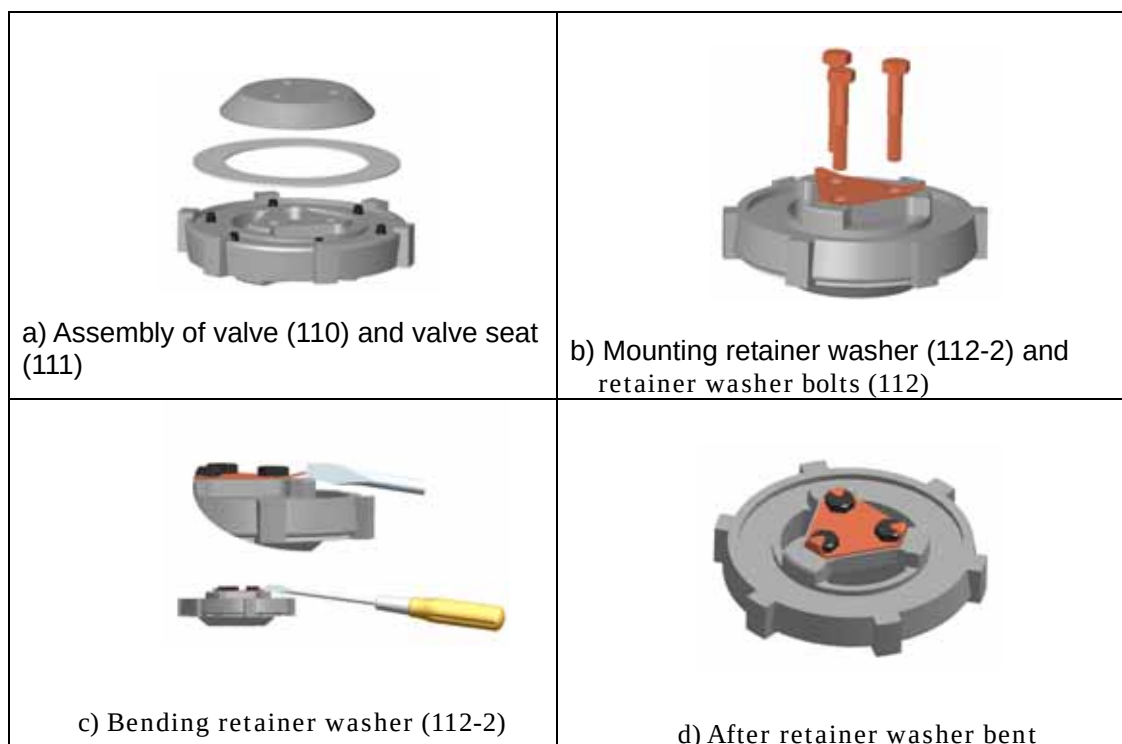


Fig.5.25 Reassembly of Discharge Valve Assembly

5.12 Head Cover (49 & 50) Reassembly

- a. Smear lubricating oil on the head cover gasket (51) and mount the gasket on the valve plate (73). Mounting the head cover should be carried out in the reverse order of

disassembly starting with the lower head covers. Screw the safety bolt into the head or fully thread bolts into the two tapered holes of the crankcase to prevent the head cover gasket (51) from slipping out of position while assuring work safety.

- b. Mount the safety head spring (117) on the discharge valve cage (109).
- c. Mount the head cover carefully while confirming that the safety head spring does not drop. Confirm that the safety head springs mate into the pits on the head cover, then mount two head cover bolts (52) in the head cover bolt holes at symmetrical positions.
- d. Screw the bolts (52) into the tapered holes in the top of the head cover while pressing around the bolts. When the two bolts engage the tapered holes, tighten them slightly while leaving clearance between the head cover and the valve plate.
- e. Screw all remaining bolts (52) into the remaining tapered holes and tighten in a crisscross pattern. Finally tighten to the specified torque.



Fig.5.26 Remounting Head Springs



Fig.5.27 Remounting Head Cover

5.13 Handhole Cover (45) Reassembly

[Caution] Before remounting the handhole cover, confirm the mating numbers of the connecting rods, the tightness of the filter (119), nuts and check that no hand tools or component parts have been left inside the crankcase.

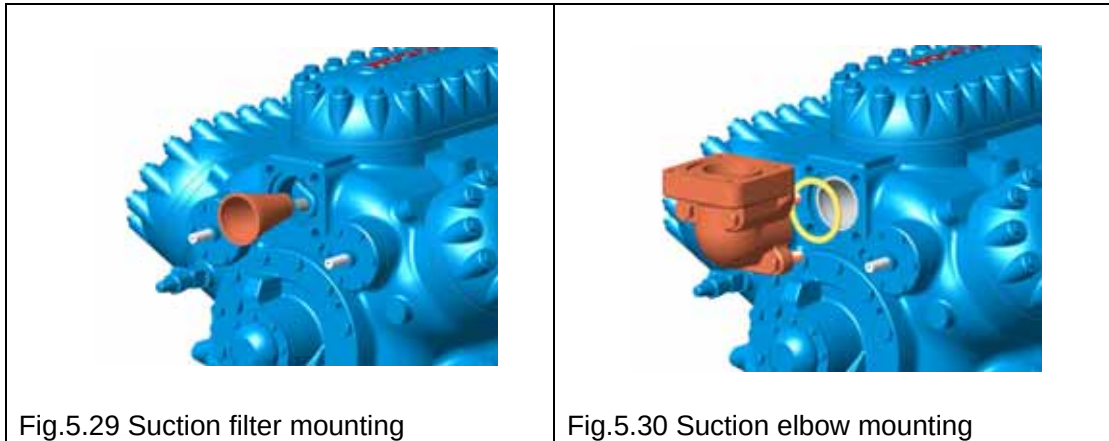
- a. Smear lubricating oil on the handhole cover gasket (47) and stick the gasket to the handhole cover.
- b. Fit the handhole cover on the crankcase and secure using two safety bolts. If the two safety bolts or fully threaded bolts are used, mounting work can easily be carried out.
- c. Screw all remaining bolts (48) into the holes and tighten them in a crisscross pattern to avoid unbalanced tightening. Finally, tighten to the specified torque.
- d. When the oil sight glass is to be renewed or dismantled, do not forget to retighten the bolts (167) to the specified torque.



Fig.5.28 Handhole Cover remounting

5.14 Suction Elbow (175) and Discharge Elbow (168) Reassembly

- a. Mount the suction filter (154) in the suction filter trap.
- b. Mount the suction elbow (175) on the crankcase and tighten the bolts (177-1 & 177-2) to the specified torque.
- c. If the discharge elbow (168) is dismantled, retighten the bolts (170-1, -2 & 170-3) to the specified torque.



5.15 Head Jacket Cover (53) Reassembly

※ Applicable for water-cooled type compressor only.

- Mount the rubber jacket cover gasket on the crankcase. Lubricating the gasket with oil is not necessary.
- Confirm the mounting position and orientation of the jacket head cover (53), then tighten the bolts (54) to the specified torque.

Fig.

5.16 External Oil Filter



- Mount the oil drain plug.
- Insert the oil filter element and confirm that the oil filter element is firmly seated.
- Check the condition of the O-ring and replace with a new one if any abnormality is observed.
- Confirm the oil flow direction indicated at the top of the flange, then tighten the bolts. Care must be taken not to mount the filter element backwards, otherwise clogging of the filter element, malfunction of the clogging indicator or damage to the filter element will result.
- Mount the locking bolt and tighten the bolts to the specified torque.

6. Parts Inspection and Replacement Standards



- MYCOM genuine parts must always be used as replacements. If bogus parts or the like manufactured by third parties are used in the compressor instead of genuine parts, malfunctions, electric shock and serious accidents such as personal injury may result.

6.1 Compressor Oil Replacement

- In addition to daily inspection of the quantity, contamination level, color, etc. of the oil, an oil sample should be extracted every six months for analysis by the manufacturer of the oil.
- If the lubricating oil becomes extremely dirty, do not hesitate to replace it. If the oil is light coffee brown in hue (ASTM-6 or more), replace the oil immediately.
- Check the oil trap in the bottom of the crankcase periodically for metallic residue. If any metal particles are found, determine the source. Residue found in the trap may include dirt from the suction piping or particles from the frictional parts of the compressor.
- If foreign matter finds its way into the compressor, the moving parts of the system may be subject to premature wear or damage. It is, therefore, vital that all precautions be taken to prevent contamination and to maintain clean oil.
- If the oil filter mounted inside the crankcase becomes dirty, carry out cleaning immediately.
- Check the external oil filter often to identify dirt at an early stage. If dirty is found in the external oil filter, replacement should be carried out as soon as possible because this filter is made of fine mesh and is prone to clogging.
- When taking an oil sample to be forwarded to the oil manufacturer for analysis, judgment should be made referring to the following standards

1. Oil Analysis

In general, analysis of the oil should be carried out to evaluate the characteristics indicated below. Normally, a 200ml (200cc) sample in a clear plastic or glass bottle is sufficient for analysis.

- ①Hue (ASTM)
- ②Kinematic viscosity [mm²/S/40°C (CST/40°C)]
- ③Total oxidation [mgKOH/g]
- ④Moisture content [ppm or vol/s]
- ⑤Millipore filter [mg/100ml]

Note: Analysis of ①~⑤above is considered routine. In addition to the above, it is also recommended that the following be checked.

- ⑥Fluidity [°C]
- ⑦Wax fog point [°C]
- ⑧Metallic contents [WC、 ppm]

2. Oil Renewal Standards

Oil should be replaced based on the following criteria:

- ①Hue : ASTM 6 or above
- ②Kinematic viscosity : Above new oil viscosity level $\pm 15\%$
- ③Total oxidation : 0.3 (mgKOH/g) or more

- ④Moisture content : 100ppm or more
- ⑤Millipore filter : 15 (mg/100ml) or more

Deterioration of the oil can generally be determined by the values of ① through ③ above. If, the moisture content or contamination by foreign particles is high, however, the oil should be replaced immediately and the source of the moisture or foreign matter traced.

※ Abnormal moisture:

- 1) The system is not properly sealed and moisture and/or air enters the system during short-term storage, or
- 2) The original oil contains an excessive amount of moisture due to long-term storage in an improperly sealed container.

※ Abnormal Millipore values:

- 1) The interior of the system has not been adequately cleaned and residue inside the piping has been flushed through other components to the compressor crankcase with the refrigerant and oil, resulting in the residue mixing with the oil in the reservoir of the crankcase, or
- 2) Abnormal abrasion of moving parts is also a potential source of residue accumulation.

6.2 Suction Filter (154) and Oil Filter (119)

- a. After cleaning the wire mesh, check for mesh damage and welding faults. If any faults are found, repair or replace with a new filter.
- b. If any dirt remains in the fine mesh after cleaning, blown out with compressed air. Aim the air blast in the opposite direction to fluid flow.

6.3 Crankshaft (2)

- a. **Check for abrasion of the crankshaft journals. Abnormal abrasion can initially be determined visually or by the simple touch of a finger which will reveal a difference in the height of friction and non-friction areas. Check each journal for wear using a micrometer. If measurements exceed service limits, replace the crankshaft.**

(mm)		
Measurement Point	Standard Dimension	Service Limit
Main bearing diameter	70	69.88
Thrust bearing diameter		
Crank pin diameter	60	59.88

- b. Frictional parts of the crankshaft may be damaged during disassembly work. If any scratches or scoring is found, polish using fine emery paper (#800 mesh or more) or an oil stone.
- c. Remove all plugs from the crankshaft and clean the oil ports thoroughly. Especially, the plugs themselves are liable to accumulate dirt. Wash the plugs thoroughly in kerosene and lubricate with oil repeatedly before reinstalling. If the plugs are not reinstalled, oil pressure failure and compressor seizure will result.
- d. Inspect the mechanical seal mounting face of the crankshaft for scratches or scoring and repair using a fine oil stone or fine emery paper (#800 mesh or more).

6.4 Mechanical Seal Assembly (32)

- a. Check the frictional face of the shaft seal cover side floating seat and the crankshaft

side seal ring. If the frictional face of the mechanical seal exhibits a lustrous finish, it is considered satisfactory. If not, replace the mechanical seal assembly.

- b. Check the frictional face of the floating seat for scratches or scoring and, if found replace the seal assembly with a new one as soon as possible.
- c. Two O-rings are used. If disassembly or inspection is carried out, replace these O-rings with new ones (also refer to 6.12 Gasket and O-ring together with the Appendix of this manual).

6.5. Piston (85), Piston Pin (86) and Piston Ring (89 , 90 & 100)

- a. The circumference of each piston should be inspected carefully for damage. If any damage is found, polish with fine emery paper, moving the paper perpendicular to the direction of movement during operation.

(mm)		
Measurement point	Standard Dimension	Service Limit
Piston diameter	85	84.80

- b. If the outer diameter of the piston exceeds the service limit given below, replace with a new one.
- c. The piston pin hole and the piston pin are designed as floatable in relation to each other. If the measured clearance exceeds the service limit given below, replace the piston pin or the piston, whichever shows the greater degree of wear.
- d. If the outer diameter of the piston pin exceeds the service limit given below, replace the piston pin with a new one.

(mm)		
Measurement Point	Standard Dimension	Service Limit
Clearance between piston pin hole and piston pin	0.007~0.026	0.15

(mm)		
Measurement Point	Standard Dimension	Service Limit
Piston pin outer diameter	25	24.85

- e. Piston ring wear can be determined by measuring the ring end gap. Insert the piston ring into the cylinder sleeve to a depth of approx. 3mm from the end and measure the ring end gap. If the gap exceeds the service limit, replace the ring with a new one.

(mm)		
Measurement Point	Standard Dimension	Service Limit
Piston ring end gap (1st, 2nd & bottom)	0.15~ 0.35	1.2

(mm)		
Measurement Point	Standard Dimension	Service Limit
Clearance between piston ring groove and piston ring	0.02 ~0.06	0.15

- f. If the clearance between the piston ring groove and the piston ring exceeds the service limit given above, replace the piston ring or the piston.

6.6. Connecting Rod (77)

- a. Inspect the surface of the bearing metal halves (84) on the big end of the connecting rod. If the bearing metal shows signs of pitting by foreign matter, wear or scoring of the crankshaft pin portion may result. Replace the bearing halves (84) with new ones.
- b. If the clearance between the crankshaft pin portion and the bearing halves exceeds the service limit given below, replace the bearing halves.
- c. If the clearance between the piston pin (86) and the small end pin hole of the connecting

rod exceeds the following service limit, replace the piston pin.

Measurement Point	Standard Dimension	(mm)	Measurement Point	Standard Dimension	(mm)
		Service Limit			Service Limit
Clearance between crank pin and bearing halves	0.020~0.079	0.20	Clearance between piston pin and connecting rod small end	0.017~0.054	0.15

6.7. Cylinder Sleeve (61)

- Check the seat surface of the cylinder sleeve under the suction plate valve (71). If scratches or scoring are found on the seat surface of the cylinder sleeve, polish the seat surface by lapping or replace the cylinder sleeve with a new one.
- If the seat portion of the cylinder sleeve is worn beyond the service limit given below, replace the cylinder sleeve with a new one.
- Check the inner wall of the cylinder sleeve and repair any scratches with a fine oil stone or fine emery paper. If scratches or scoring are bad, replace the cylinder sleeve with a new one.
- Check the clearance between the inner surface of the cylinder sleeve and the outer surface of the lower portion of the piston. If the clearance exceeds the following service limit, replace the cylinder sleeve with a new one.
- If the inner diameter of the cylinder sleeve exceeds the service limit given below, replace the sleeve. Measure the diameter of the cylinder sleeve at a point 10~20mm below the top of the cylinder, as this area is prone to wear the most.

Measurement Point	Standard Dimension	(mm)	Measurement Point	Standard Dimension	(mm)
		Service Limit			Service Limit
Height of seat portion of cylinder sleeve	0.5	0.2	Clearance between inner surface of cylinder sleeve and lower portion of piston	0.080~0.132	0.3
Inner diameter of cylinder sleeve	85	85.2			

6.8. Discharge Valve (110), Suction Valve (71) and Springs (72 & 116)

- The discharge plate valve (110), suction plate valve (71) and valve springs (72 & 116) must be replaced every 6,000 hours of operation. Durability of these components varies according to operating conditions but the wear limit for the valve seat portion should be less than 0.15mm from the standard thickness shown below.

(mm)

Part Name	Standard Thickness	Service Limit
Discharge plate valve (110)	1.0	0.85 (Valve seat portion)
Suction plate valve (71)		

- Even if abrasion of the plate valve seat portion is within the service limit, if scratches, scoring or other damage is found on either the external or internal circumference, this can lead to breakage due to metal fatigue, so replace the plate valve with a new one.
- The allowable wear limit for the discharge valve seat (111) and the seat portion of the valve plate (73) is 0.2mm.

6.9. Oil Pump

- If the oil pressure is too low even after throttling the oil pressure regulating valve (22) and the oil filter (119) is found not to be clogged, one possible cause is wear of the pump gear, metal and shaft. When carrying out inspection of the oil pump, detach the pump from the compressor, check the play of the pump shaft by moving the shaft in both the parallel and perpendicular directions using a pair of pliers or nippers. If shaft play is felt,

replace the bearing housing assembly with a new one.

- b. Even if the oil pump shaft rotates smoothly, it is possible that the surface of the gear portion or the bearing shaft portion of the oil pump is worn out or the pump has been contaminated by foreign matter. Replace the bearing housing assembly with a new one.

6.10. Main Bushing (12) and Thrust Washers (29-1 & 29-2)

- a. If the clearance between the main bushing and the crankshaft exceeds the following service limit, replace the main bearing and thrust washer.
- b. If the inner diameter of the main bushing exceeds the service limit given below, replace the bushing with a new one.
- c. If the surface of the thrust washer seems slightly rough but no problem is evident, polish the surface using fine emery paper or an oil stone. If the thickness of the thrust washer is less than the service limit, replace with a new one.

(mm)			(mm)		
Measurement Point	Standard Dimension	Service Limit	Measurement Point	Standard Dimension	Service Limit
Clearance between main bushing and crankshaft	0.04~0.139	0.3	Inner diameter of bearing	70	70.15
			Thrust washer	3.0	2.8

6.11. Springs (69, 72, 116 & 142)

- a. Measure the free length of each spring. If the free length of the spring is 10% or more shorter than the standard value given below, replace the spring with a new one.
- b. The discharge valve spring (116) and the suction valve spring (72) should be replaced every 6,000 hours of operation. In addition, if any abnormality is found such as breakage or deformation, etc., do not hesitate to replace the spring(s) with new ones.

(mm)	
Name of Spring	Standard dimensions (Free length x coil dia. x outer dia.)
Discharge valve spring (116)	
Suction valve spring (72)	10 x t0.2 x ϕ 6 (max. dia.)
Unloader lift pin spring (69)	10 x ϕ 0.4 x ϕ 6 (max. dia.)
Unloader device spring (142)	72 x ϕ 2.8 x ϕ 32.3

6.12. Gaskets and O-rings

- a. After disassembly and inspection have been carried out, replace the gaskets with new ones.
- b. Although the O-rings used are made of synthetic rubber, they may deteriorate due to exposure to refrigerant and oil over a long period of time. Periodic replacement is required when disassembly and inspection is carried out, regardless of appearance.

[Caution] The material of gaskets and O-rings depends on the kind of refrigerant used, therefore do not forget to designate the refrigerant when ordering.

6.13. Oil Filter Element (Externally mounted type)

The oil filter element cannot be refreshed by washing. If the element is extremely dirty, deformed or worn out, replace with a new one.

Appendix No.1 Table of Component Parts

Table of Component Parts of MYCOM Model 6HK Reciprocating Compressor

No.	Name	Material	Quantity	Remarks
			6HK	
1	Crankcase	FC300	1	
2	Crankshaft	SFCM930S	1	
2-1	Retainer Drive pin	SK5	1	Φ3×L8, Spring pin
4	Pulley key	S45C	1	15×9.7×60
5	Flat washer, pulley	SGD-B	1	Common use with Model K
6	Spring washer, pulley	SWRH57	1	For M20
7	Set bolt, pulley	~	1	M20×L50-10.9T
8	Housing, bearing	FC250	1	Combined with oil pump.
9	Gasket, bearing housing	Non-asbestos	1	
10-1	Locking bolt, bearing housing (1)	~	10	M12×L45-10.9T
10-2	Locking bolt, bearing housing (2)	~	2	M12×L50-10.9T
10-3	Spring washer, bearing housing	SWRH57	2	For M12
12	Main bushing	WJ1 (F1)	2	Common use with Model K
13	Locking pin, thrust washer	SK5	4	Φ4×L8, Spring pin
20	Oil charge valve (oil supply/drain)	~	1	R3/8×Rc 3/8
22	Oil pressure regulating valve (Assy.)	~	1	Common use with Model C R1/2
26	Shaft seal cover	FC250	1	
27	Gasket, shaft seal cover	Non-asbestos	1	
28	Bolt, shaft seal cover	~	8	M10×L45-10.9T
29-1	Thrust washer (1)	CAC603 (F100)	1	Φ100
29-2	Thrust washer (2)	CAC603 (F100)	1	Φ93, Common use with Model K
32	Mechanical seal assembly	~	1	
45	Handhole cover	FC300	2	With sight glass
47	Gasket, handhole cover	Non-asbestos	2	
48	Bolt, handhole cover	~	24	M12×L55-10.9T
49	Head cover (air-cooled) ※1	FC300	3	Exclusive use for air-cooled type
50	Head cover (water cooled) ※1	FC300	3	Exclusive use for water-cooled type
51	Gasket, head cover	Non-asbestos	3	
52	Bolt, head cover	~	54	M12×L100-10.9T
53	Head jacket cover ※1	FC250	3	Exclusive use for water-cooled type
54	Bolt, head jacket cover ※1	SUS	24	M10×L30, Exclusive use for water-cooled type
55	Gasket, head jacket cover ※1	Natural rubber	3	Exclusive use for water-cooled type
61	Cylinder sleeve	FC250 (Boron cast iron)	6	Common use with Model K
61-1	Positioning pin, cylinder sleeve	SK5	6	Φ3×L8, Spring pin
62L	Cam ring (Left downward slanted)	FC250	2	Common use with Model K
63R	Cam ring (right downward slanted)	FC250	2	Common use with Model K
65	Retaining ring, unloader	SK5	4	Common use with Model K
68	Lift pin, unloader	S50C-D	16	Common use with Model K
69	Spring, unloader lift pin	SWC	16	Common use with Model A
70	Stop ring, unloader lift pin (E-ring)	S60CM	16	E4
71	Suction plate valve	SUS420J2-M0	6	Common use with Model K
72	Spring, suction plate valve	SK5	24	Common use with Model C
73	Valve plate	FCD450	3	

No.	Name	Material	Quantity	Remarks
			6HK	
73-1	Bolt, valve plate (1)	~	6	M12×L30-10.9T
73-2	Bolt, valve plate (2)	~	3	M10×L25-10.9T
73-3	Positioning pin, valve plate	S45C	6	Φ6×L18, Parallel pin Class B
73-4	Gasket, valve plate	Non-asbestos	3	
74	Cage guide, discharge valve	FC300	6	
75	Bolt, discharge valve cage guide	~	30	M6×L30-10.9T
75-1	Positioning pin, cage guide	S45C	12	Φ6×L18, Parallel pin
77	Connecting rod	T31-T6	6	Common use with Model K
78	Bolt, connecting rod	SCM435	12	M10×L81, Common use with Model K
79	Washer, connecting rod	S45C	12	For M10, Common use with Model K
80	Nut, connecting rod	S45C	24	For M10, Common use with Model K
84	Bearing halves, connecting rod	A17X	12	
85	Piston	AC8A	6	Common use with Model K
86	Piston pin	SCr420H	6	Φ25×L73, Common use with Model K
87	Lock spring, piston pin	SK5	12	H25, Common use with Model K
89	Piston ring (top) (FC-P) *2	FC250	6	For CO₂ and HFC Common use with Model K
89	Piston ring (top) (GA-P) *2	LBC-2	6	NH₃ , Common use with Model K
90	Piston ring (2 nd) (FC-P)	FC250	6	Common use with Model K
100	Oil ring (Bottom) (FC-CO)	FC250	6	Common use with Model K
109	Cage, discharge valve	FC300	6	
110	Discharge plate valve	SUS420J2-M0	6	Common use with Model K
111	Seat, discharge valve	SNC631	6	Common use with Model K
112	Bolt, discharge valve seat	~	18	M6×L35-10.9T
112-2	Retainer washer, discharge valve	SPCC	6	t=0.9, Common use with Model K
115	Positioning pin, discharge valve seat	S45C	6	Φ6×L18, Parallel pin Class B
116	Spring, discharge valve	SK5	36	Common use with Model C
117	Spring, safety head	SUP60TW	6	Common use with Model WA
118	Hex head nipple	S25C	1	R1/2×R1/2
119	Oil filter	SUS316	1	#200, Common use with Model C
135	Push rod, unloader (1)	AC2B-F	1	Common use with Model K
136	Push rod, unloader (2)	AC2B-F	1	Common use with Model K
142	Spring, unloader device	SWC	2	Common use with Model L
143	Washer, unloader push rod	S25C	2	Common use with Model C
144	Stop ring, unloader push rod	SK5	2	S20, Common use with Model C
145	Unloader piston	FC250	2	
146	Unloader cover	~	2	(Built-in solenoid valve)
147	Gasket, unloader cover	Non-asbestos	2	
149	Bolt, unloader cover	~	12	M8×L40-10.9T
149-1	Washer, unloader cover locking bolt	S25C	12	For M8
154	Suction filter	SUS316	1	
164	Glass, oil sight	~	2	
165	O-ring, oil sight glass	Chloroprene Rubber (CR)	2	Oil sight glass Accessory
167	Bolt, oil sight glass gland	~	6	M6×L15-12.9T, (Hexhead hole)
168	Discharge elbow	FC300	1	
169	Gasket, discharge elbow	Non-asbestos	1	

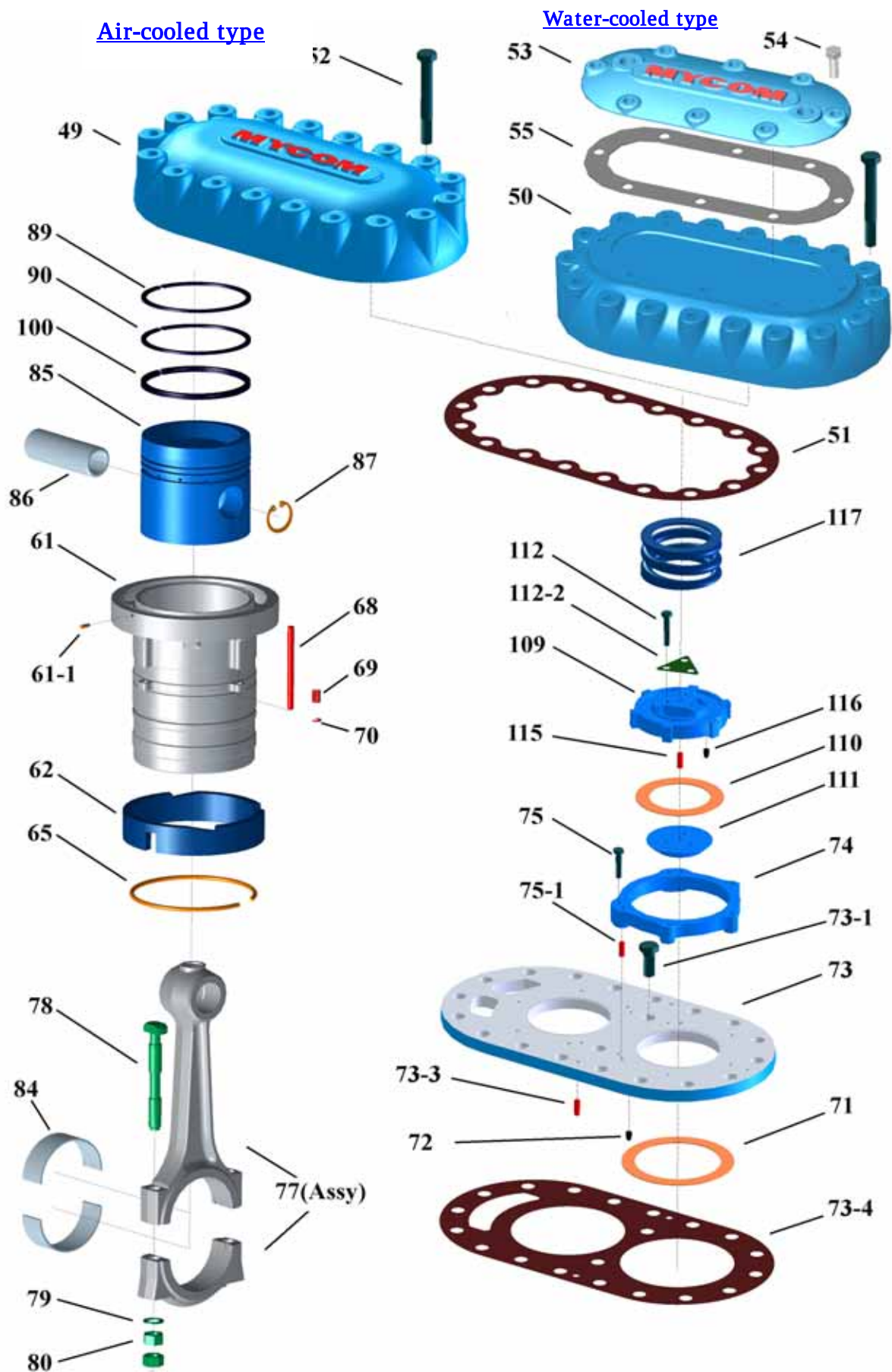
No.	Name	Material	Quantity	Remarks
			6HK	
170-1	Bolt, discharge elbow (1)	~	2	M12×L130-10.9T
170-2	Bolt, discharge elbow (2)	~	3	M12×L55-10.9T
170-3	Bolt, discharge elbow (3)	~	1	M12×L50-12.9T (Hex head)
171	Gas (air) purge valve	~	1	R1/4×Rc3/8
172	Gasket, discharge companion flange	Non-asbestos	1	50A
173-1	Companion flange, discharge	S25C	1	RTK 50A
174	Bolt, discharge companion flange	~	4	M16×L45-10.9T
175	Suction elbow	FC250	1	
176	Gasket, suction elbow	Non-asbestos	1	Common use with Model K
177-1	Bolt, suction elbow (1)	~	2	M16×L135-10.9T
177-2	Bolt, suction elbow (2)	~	2	M16×L55-10.9T
182-1	Companion flange, suction	S25C	1	RTK 65A
183	Bolt, suction companion flange	~	4	M18×L50-10.9T
184	Gasket, suction companion flange	Non-asbestos	1	65A
208	Eye bolt, hanger	SF390	2	M12
245	Oil heater	~	1	
245-1	Bushing, oil heater (hexhead)	S25C	1	AB-25×25, Common use with Model A
305	Pressure equalizing piping assembly	~	1	
-	Oil filter, external	~	1	

- ※ 1 The head cover component differs depending on whether the unit is air-cooled or water-cooled.
 Air-cooled type: Air-cooled type head cover
 Water-cooled type: Water-cooled head cover consisting of water jacket cover, jacket cover gasket and locking bolts.

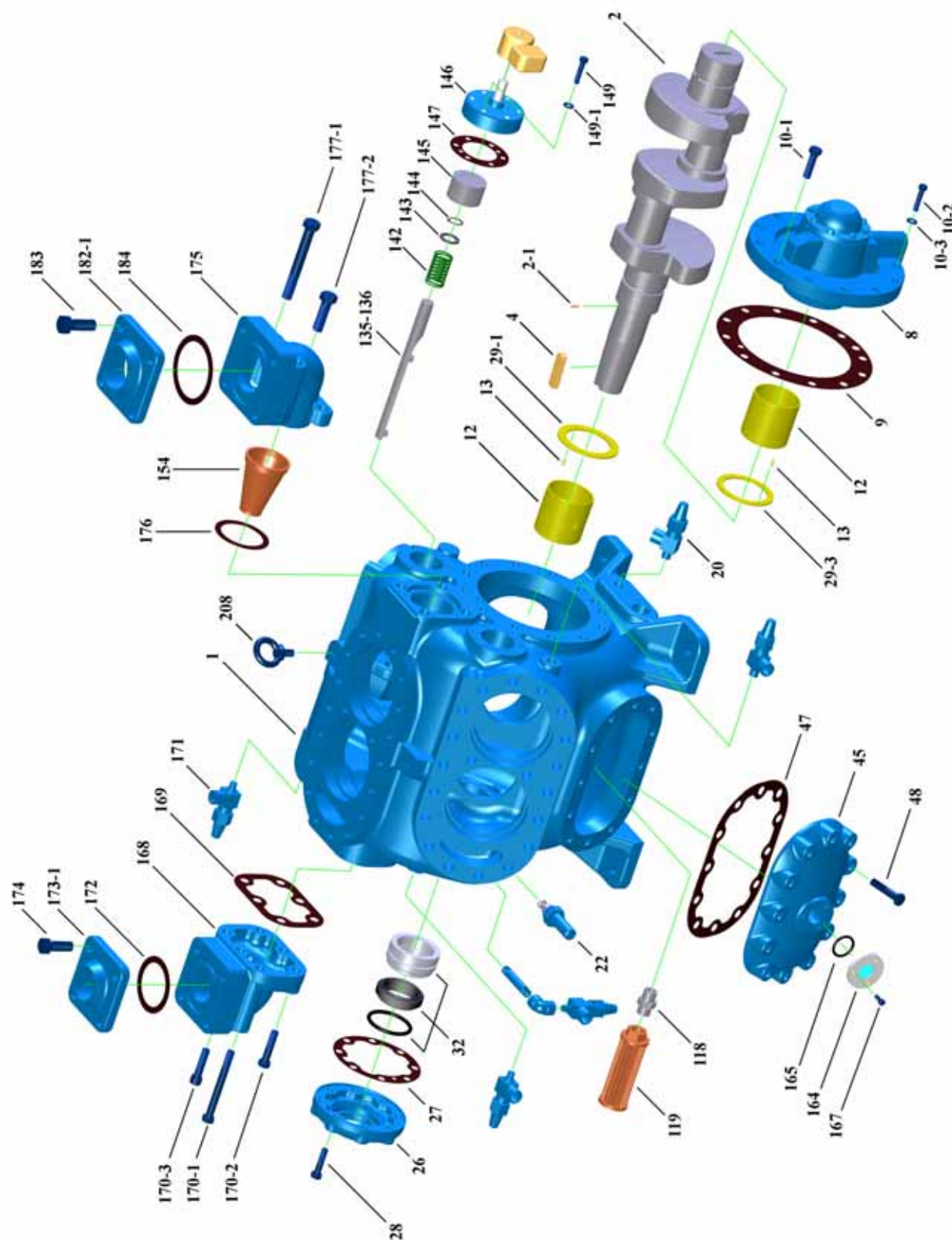
- ※ 2 For compressors using CO₂, HFC or NH₃ refrigerant, the material of the piston rings differs as follows:

Part No.	Name		CO ₂ /HFC	NH ₃
89	Piston ring (Top)	1 st	FC-P (Black)	GA-P (Yellow)
90	Piston ring (2 nd)	2 nd	FC-P	
100	Oil ring (Bottom)	3 rd	FC-CO	

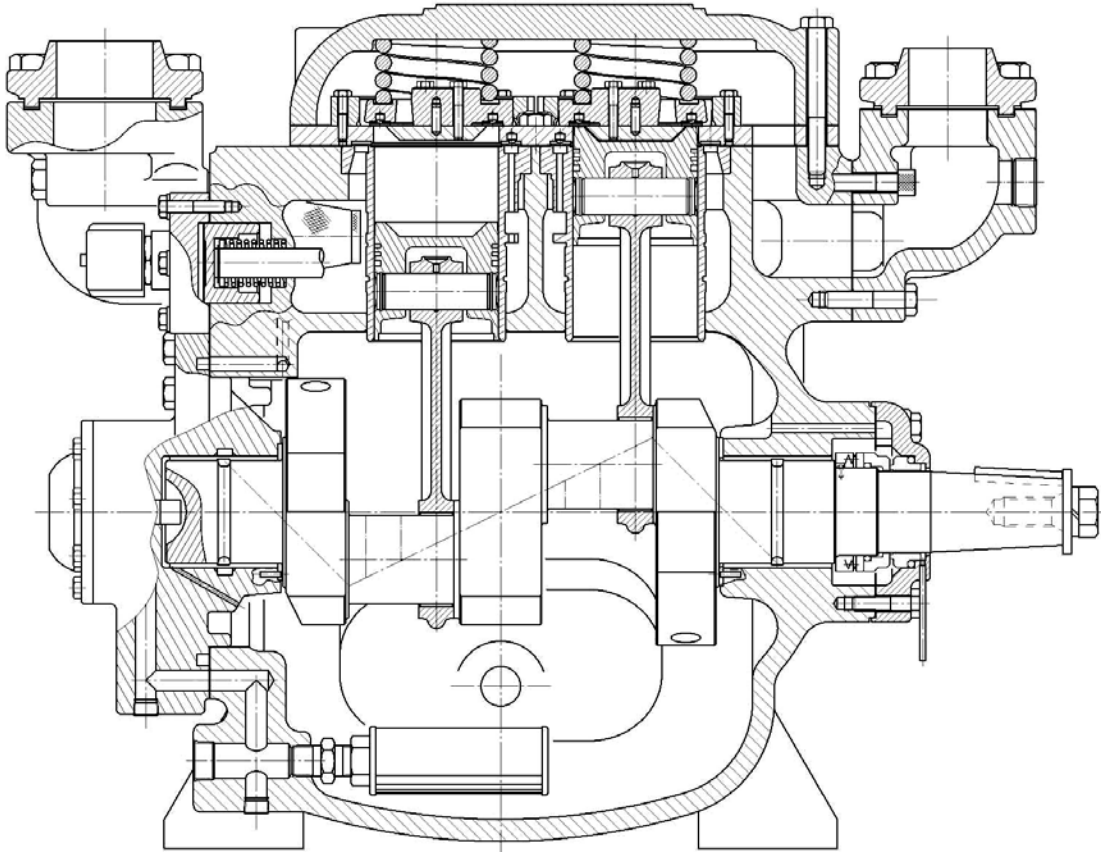
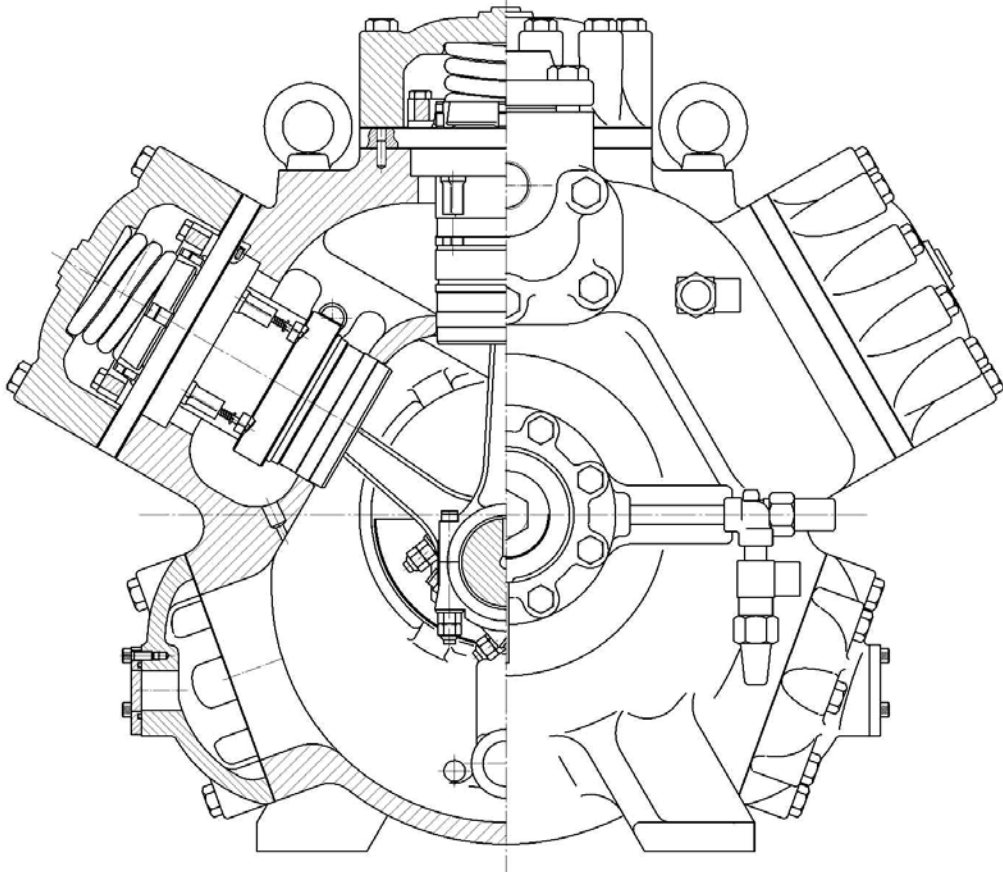
Appendix No.2 Exploded View of Cylinder portion, Model 6HK Reciprocating Compressor



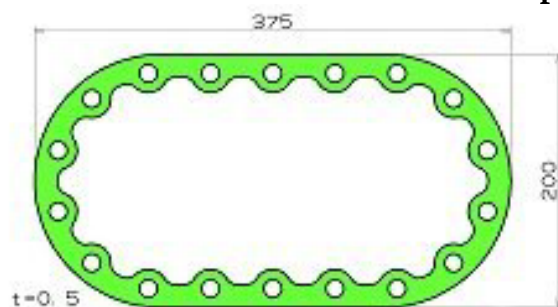
Appendix No.3 Exploded View of Model 6HK Compressor



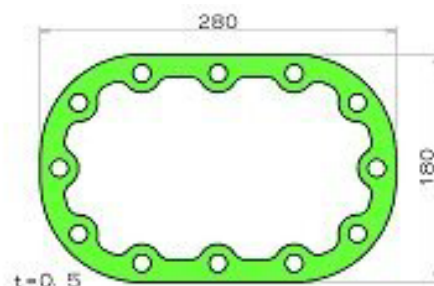
Appendix No.4 Sectional View of Model 6HK Compressor



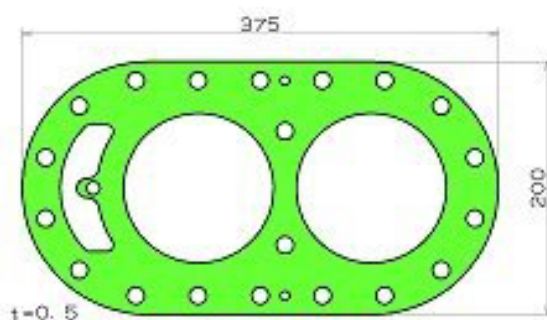
Appendix 5. List of Gaskets for Model 6HK Compressor



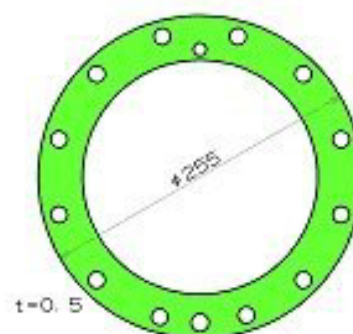
Gasket, Head cover (51)



Gasket, Hand hole cover (47)



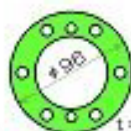
Gasket, Valve plate (73-4)



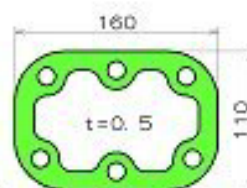
Gasket, Bearing housing (9)



Gasket, Shaft seal cover (27)



Gasket, Unloader cover (147)



Gasket, Discharge elbow (169)



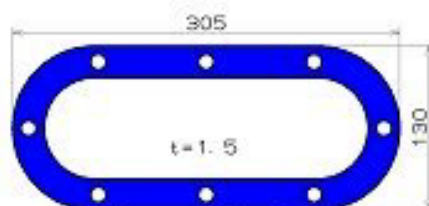
Gasket, Suction elbow (176)



Gasket, Companion flange 65A (184)

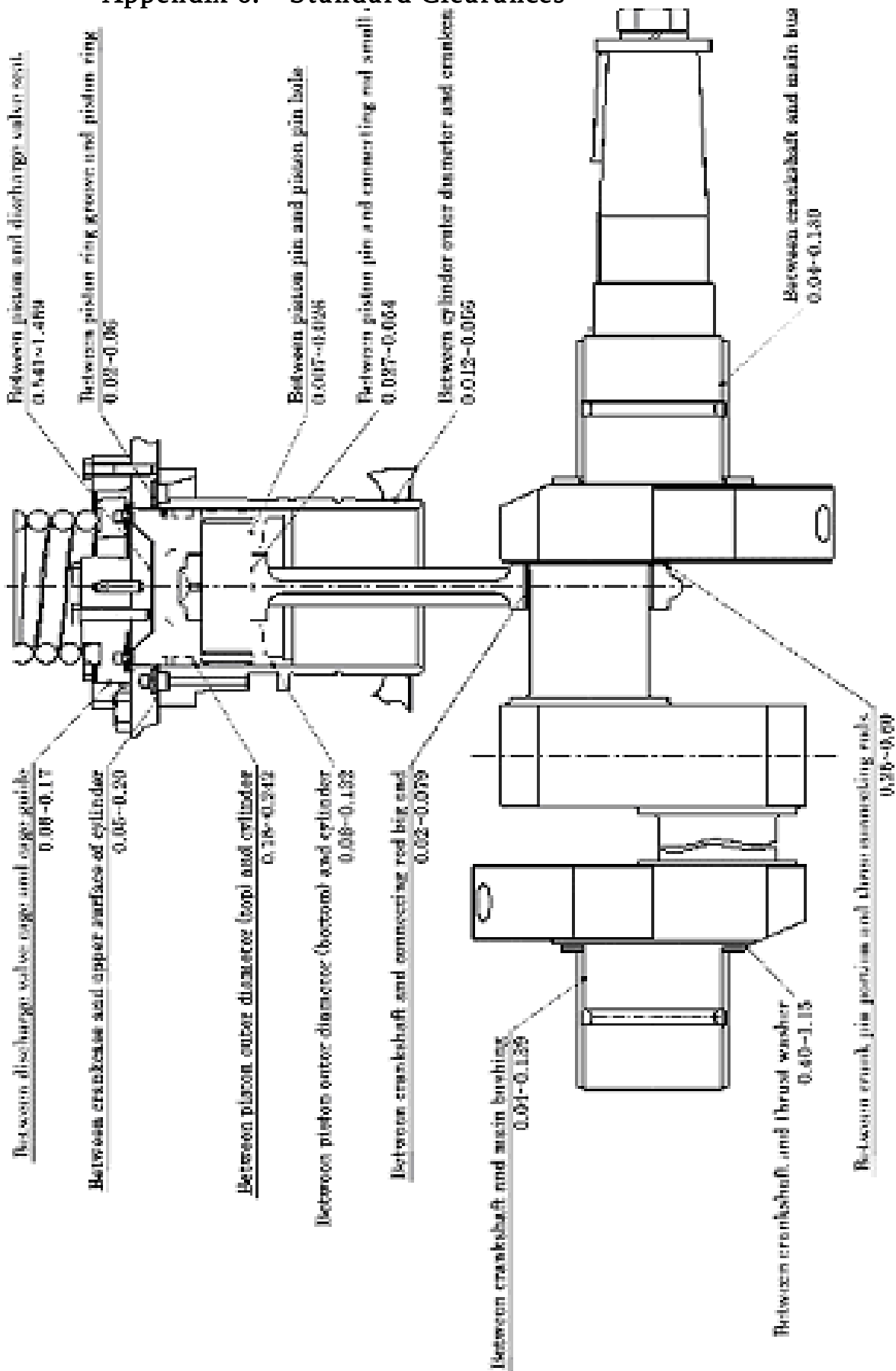


Gasket, Companion flange 50A (172)



Gasket, Jacket head cover (rubber) (55)

Appendix 6. Standard Clearances



List of Standard Clearances

Appendix No.7 Tightening Torque Values of Bolts & Nuts for Model 6HK Compressor

No.	Size of Bolt	Tensile strength or (Material)	Qty. used 6HK	Name of Bolt (Part No.)	Tightening Torque N•m (kg • cm)
1	M20 x L50	10.9	1	Pulley set bolt	160 (1600)
2	M18 x L50		4	Bolt, suction companion flange (184)	140 (1400)
3	M16 x L135		2	Bolt, suction elbow (No.1) (177-1)	120 (1200)
4	M16 x L55		2	Bolt, suction elbow (No.2) (177-2)	
5	M16 x L45		4	Bolt, discharge companion flange (174)	
6	M12 x L130		2	Bolt, discharge elbow (No.1) (170-1)	80 (800)
7	M12 x L100		54	Bolt, head cover (52)	
8	M12 x L55		24	Bolt, hand hole cover (48)	
9	M12 x L50		3	Bolt, discharge elbow (No.2) (170-2)	
10	M12 x L45		2	Bolt, bearing housing (No.2) (10-2)	
11	M12 x L30		10	Bolt, bearing housing (No.1) (10-1)	
12	M12x L50 With hex-head hole	12.9	6	Bolt, valve plate (1) (73-1)	100 (1000)
13	M12xL35 With hex-head hole	12.9	1	Bolt, discharge elbow (No.3) (170-3)	
14	M12 Eye bolt	SF390	4	Bolt, balance weight	
15	M10 x L81 (Special)	SCM435	2	Eye bolt, hanger (208)	-
16	M10 x L45	10.9	12	Bolt, connecting rod (78)	-
17	M10 x L30		8	Bolt, shaft seal cover (28)	60 (600)
18	M10 x L25		24	Bolt, head jacket cover (54)	
19	M10 Nut (Special)	S45C	3	Bolt, valve plate (No.2)	
20	M8 x L40		12	Nut, connecting rod (No.1) (80)	40 (400)
21	M6 x L30	10.9	12	Nut, connecting rod (No.2) (80)	40 (400)
22	M6 x L35		12	Bolt, unloader cover (149)	40 (400)
23	M6x L15 With hex-head hole	12.9	30	Bolt, discharge valve cage guide (75)	15 (150)
			18	Bolt, discharge valve seat (112)	
			6	Bolt, oil sight glass gland (167)	10 (100)



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